

Introduction to Bayesian Statistics

Part 6 Linear Mixed Models

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This lecture

- Random grouping factors
- Random effects
- Fixed & random effects
 - Random intercepts
 - Random intercepts and slopes
- Multiple random grouping factors
- Continuous correlation structures
- Summary

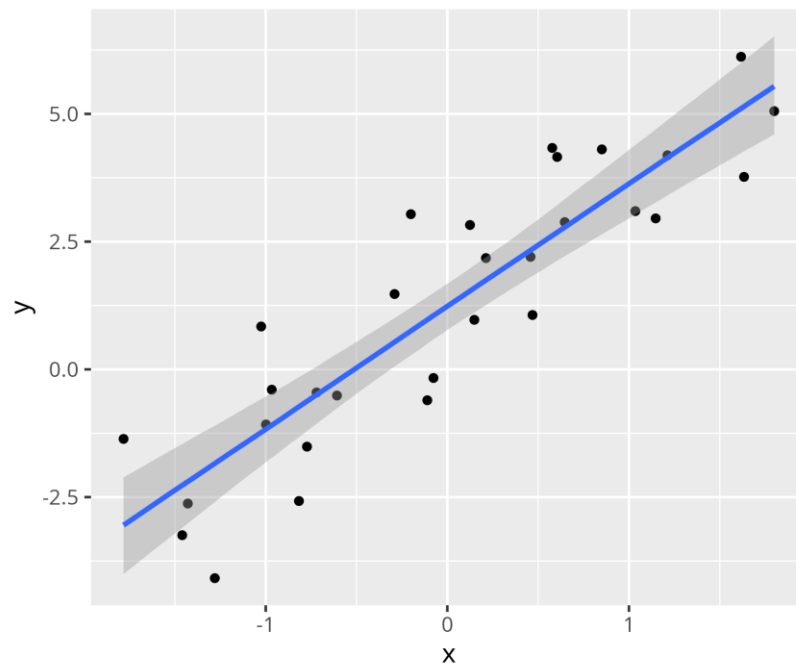
Assumptions for linear models

1. Linearity: Trend of y follows
linear prediction model $\mu(x) = a + b \cdot x$

~~**2. Independence** of residuals:
not correlated (e.g. space or time)~~

3. Equal variance of residuals:
across whole range of x

4. Normality of residuals:
 $\varepsilon \sim \text{Normal}(0, \sigma)$

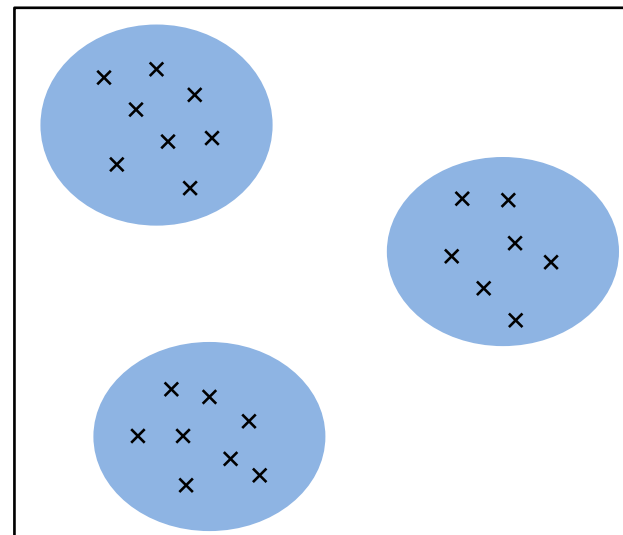


Assumptions for linear models

When are residuals not independent ?

Study design: N observations have been **sampled in groups**

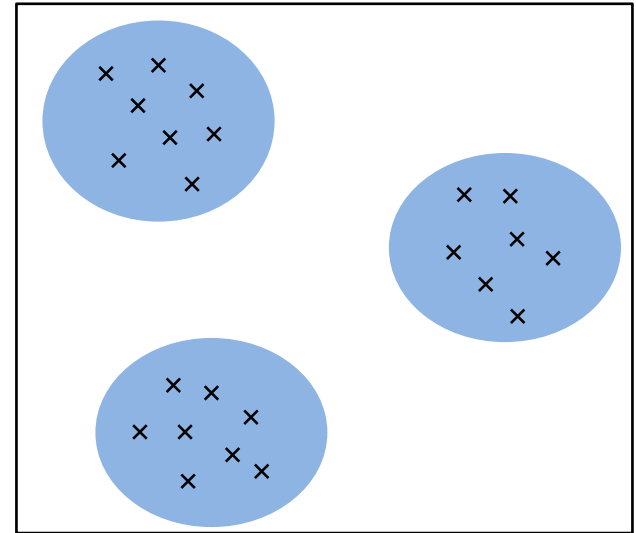
- There is a categorical predictor with M levels ($M < N$)
- Each level / group contains multiple observations, these are no independent replicates !
- Maybe structural (or random) differences between (unobserved) conditions of these groups
- Still want to make **inference on whole population**



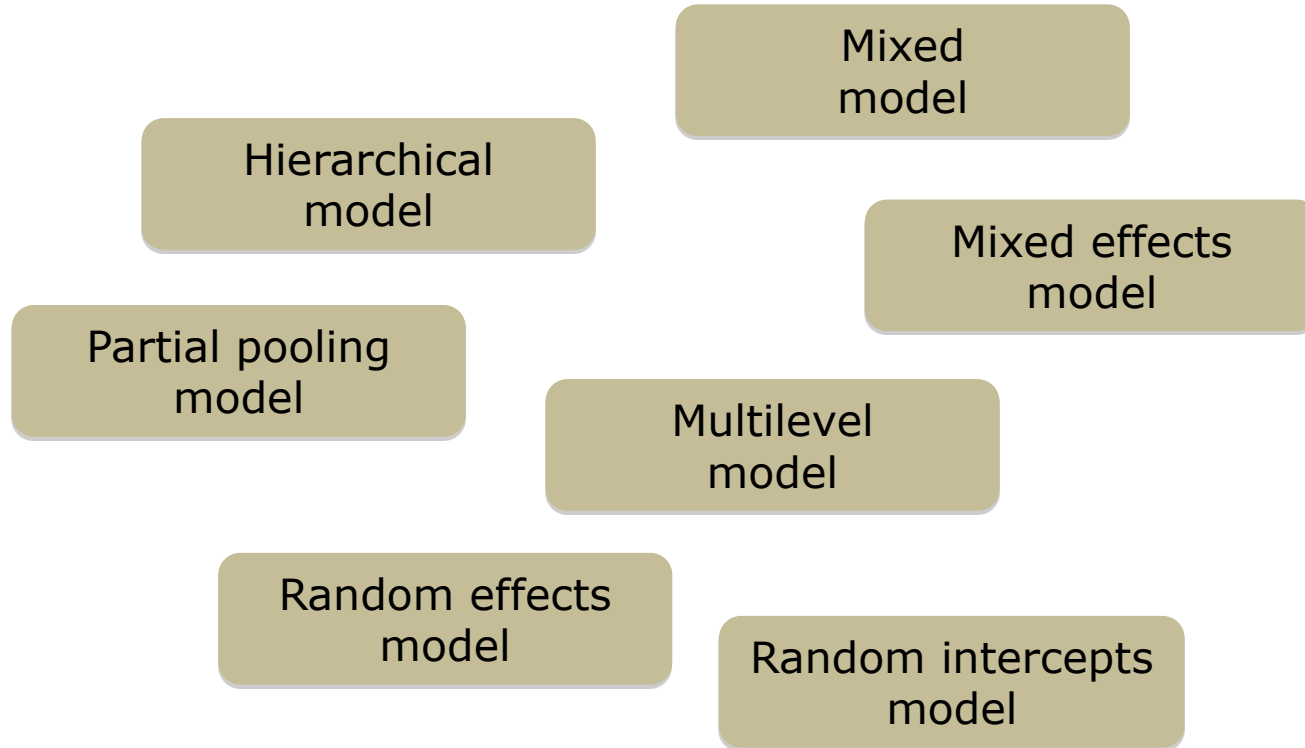
Random grouping factors

Examples:

- Site: Data collected at multiple locations
- Year: Data collected ~same time in different years
- Observer: Data collected by different researchers
- Individual: Multiple measurements on the same test object
- Unit: Multiple experimental units like chambers / tanks
- Species: Data on higher level (e.g. family) contains multiple species



Word salad !!



Random effects

Example

- Measured weight of $N = 200$ viper snakes
- Data collected on $M = 9$ different sites across France
- Also measured length, but we're not using it for now

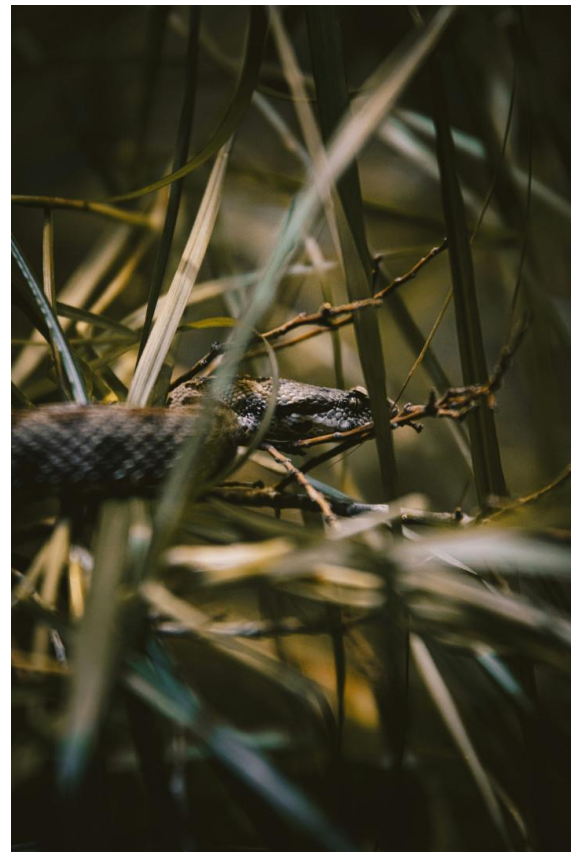
Q: What is the correct statistical model for species mean weight ?

`brm( weight ~ 1 ) ?`

→ No, observations not independent

`brm( weight ~ site ) ?`

→ No, does not estimate overall weight (just site-specific means)



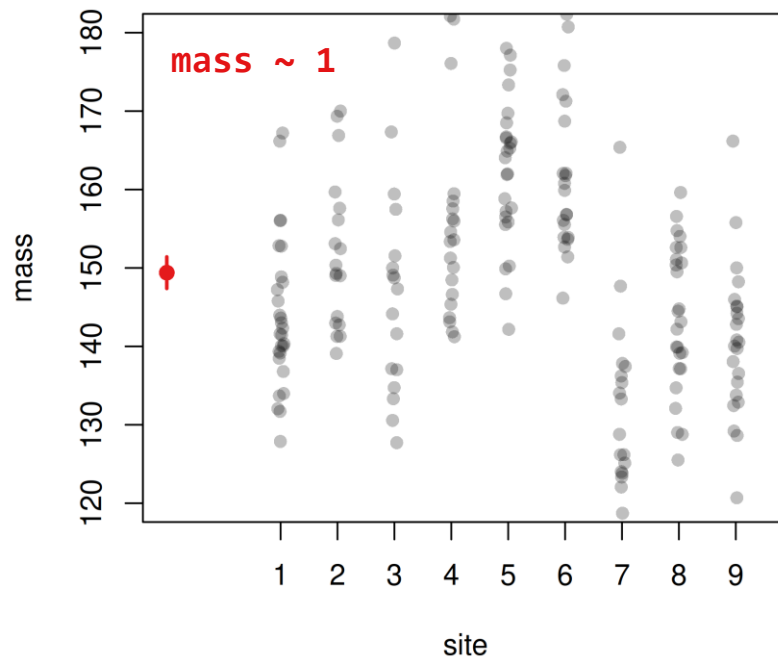
(1) Complete pooling

Model: $y_i = a + \varepsilon_i$ $i = 1 \dots N$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

Priors: $a \sim \text{Normal}(150, 10)$

- Ignore categorical predictor *site*
- Information is completely pooled across all levels
- Fit joint intercept a only
- a is given a prior



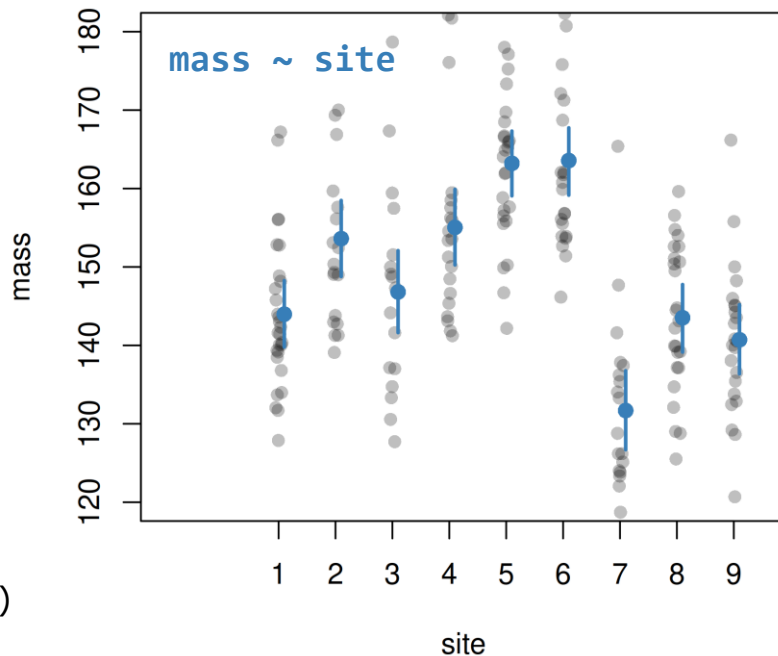
(2) No pooling

Model: $y_i = a_{site(i)} + \varepsilon_i \quad i = 1 \dots N$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

Priors: $a_j \sim \text{Normal}(150, 10) \quad j = 1 \dots M$

- Include categorical predictor *site* (**ANOVA**)
- No information is shared across levels
- Fit independent intercepts $a_1 \dots a_M$
- Each a_j is given a prior
(here they are identical, but could also be different)



(3) Partial pooling

Model: $y_i = a_{\text{site}(i)} + \varepsilon_i \quad i = 1 \dots N$

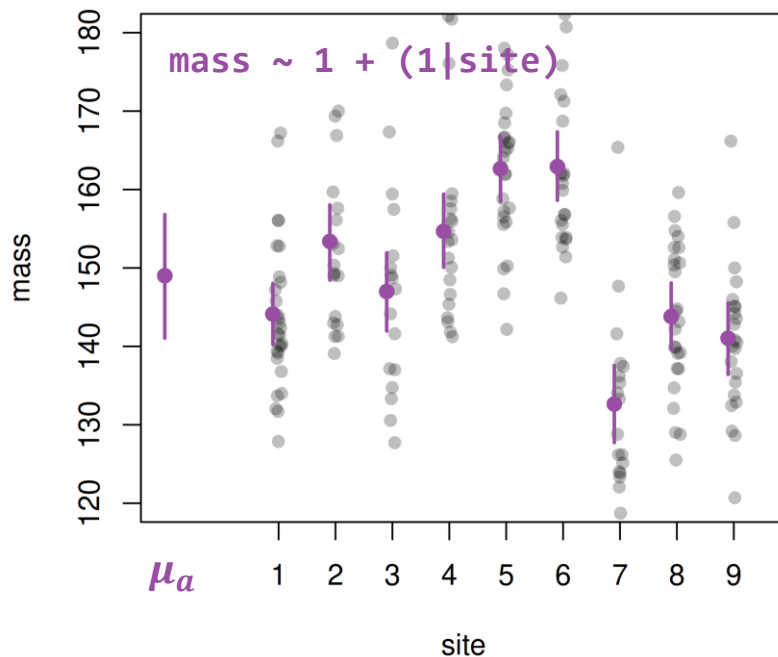
$\varepsilon_i \sim \text{Normal}(0, \sigma)$

$a_j \sim \text{Normal}(\mu_a, \sigma_a) \quad j = 1 \dots M$

Priors: $\mu_a \sim \text{Normal}(150, 10)$

$\sigma_a \sim \text{Exponential}(1/10)$

- Replace prior distributions by a joint distribution of the site-specific intercepts $a_1 \dots a_M$
- Their mean μ_a and sdev σ_a are parameters and are estimated with all others by MCMC
- These “hyperparameters” μ_a, σ_a are given priors



(3) Partial pooling

Model: $y_i = \mu_a + \delta_{\text{site}(i)} + \varepsilon_i \quad i = 1 \dots N$

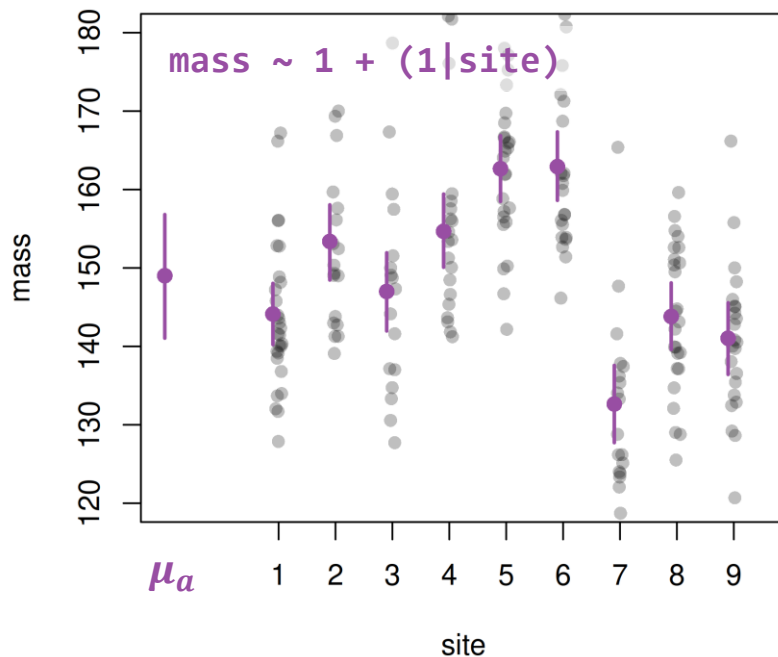
$\varepsilon_i \sim \text{Normal}(0, \sigma)$

$\delta_j \sim \text{Normal}(0, \sigma_a) \quad j = 1 \dots M$

Priors: $\mu_a \sim \text{Normal}(150, 10)$

$\sigma_a \sim \text{Exponential}(1/10)$

- Replaced $a_j = \mu_a + \delta_j$
- Effects $\delta_j = a_j - \mu_a$ describe each site's deviation from overall mean μ_a
- Identical model, just rearrangement of coefficients



Comparison

vs. complete pooling $\text{mass} \sim 1$

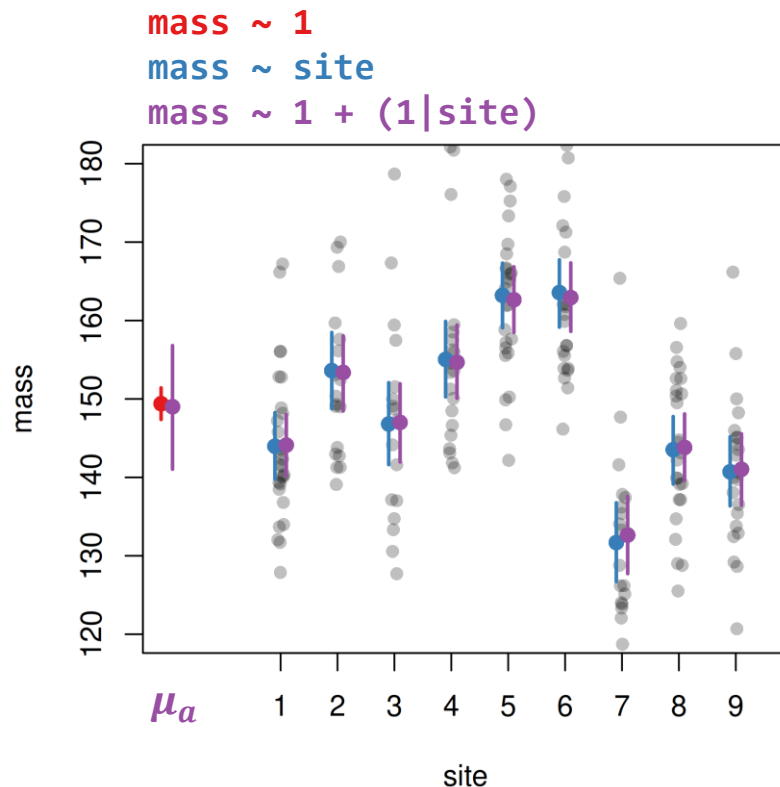
Uncertainty of μ_a in $\text{mass} \sim 1$ is smaller.
Wrongly assumes $N = 200$ independent obs.

Uncertainty of μ_a in $\text{mass} \sim 1 + (1|\text{site})$ correct.
Informed by $M = 9$ sites.

vs. no pooling $\text{mass} \sim \text{site}$

Site-specific means a_j of $\text{mass} \sim 1 + (1|\text{site})$
closer to overall mean μ_a than in $\text{mass} \sim \text{site}$

„Shrinkage“ of parameters



Random effects

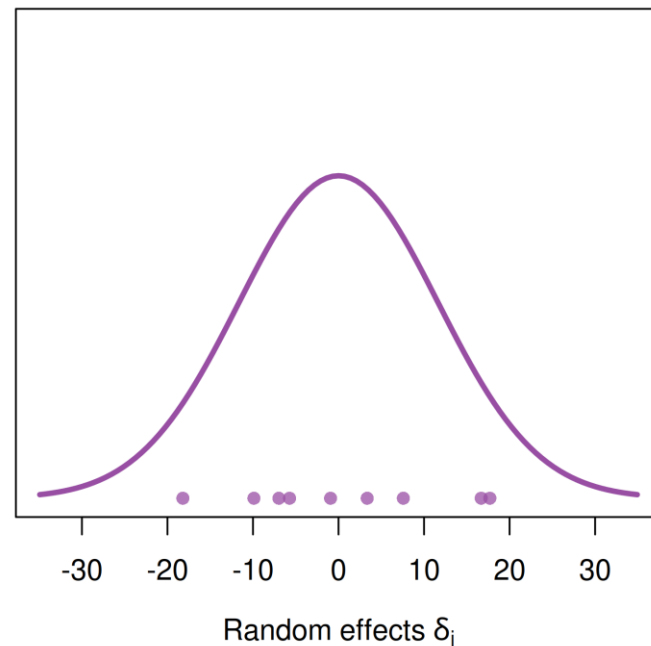
Random effects $\delta_j = a_j - \mu_a$ describe each site's deviation from overall mean μ_a with sdev σ_a

Assumption: normally distributed

(Bayesian 3d printer: could also be other distribution, e.g. positive lognormal distribution for a_j)

Hyperparameters μ_a, σ_a informed by all 9 sites

- Some information is shared / pooled
- Borrowing strength
- For unbalanced designs this improves prediction in levels with few observations.



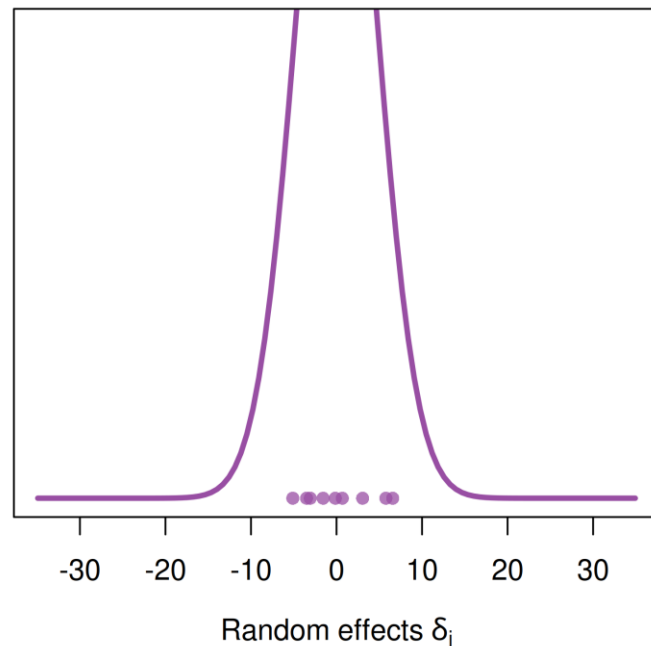
Random effects

Random effects $\delta_j = a_j - \mu_a$ describe each site's deviation from overall mean μ_a with sdev σ_a

This part of the likelihood $\delta_j \sim \text{Normal}(0, \sigma_a)$ would be maximized for $\sigma_a \rightarrow 0$ and all $\delta_j \rightarrow 0$

But the residuals' likelihood $y_i \sim \text{Normal}(\mu_a + \delta_{\text{site}(i)}, \sigma)$ would decrease if all δ_j were close to 0 (worse model fit)

→ Partial pooling model is a compromise between small random effects (on site-level) and good model fit (on observation level)



Variance partitioning

Overall variance of the data is estimated on different levels

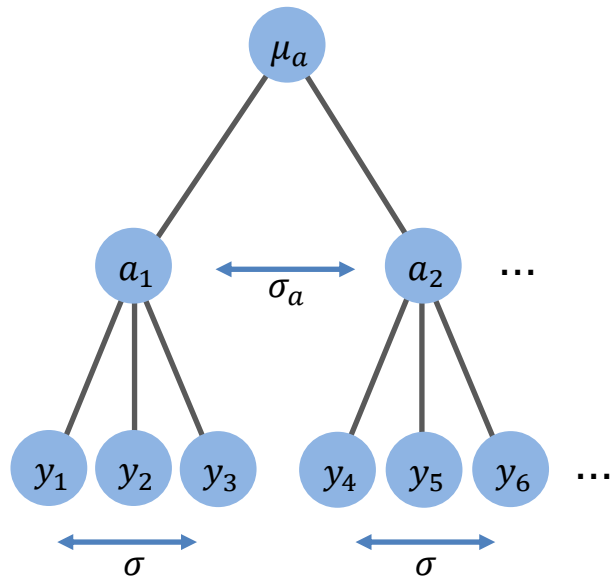
$$s_{\text{total}}^2 = s_{\text{among}}^2 + s_{\text{within}}^2$$

Site-level (among) σ_a

Explains site-specific deviation from overall mean μ_a

Observation-level (within) σ

Explains each data point's deviation from site-mean a_j
(All the randomness in the data which is not caused by site-level differences)



> brm(mass ~ 1 + (1|site), ...)

```
Family: gaussian
Links: mu = identity; sigma = identity
Formula: mass ~ 1 + (1 | site)
Data: df (Number of observations: 200)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

~site (Number of levels: 9)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
$\sigma_a \rightarrow$ sd(Intercept)	11.64	3.26	6.99	19.70	1.01	814	1353

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
$\mu_a \rightarrow$ Intercept	149.01	3.91	141.05	156.81	1.00	777	1038

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
$\sigma \rightarrow$ sigma	11.02	0.57	9.97	12.20	1.00	2527	2566

Fixed and random effects

Fixed & random effects

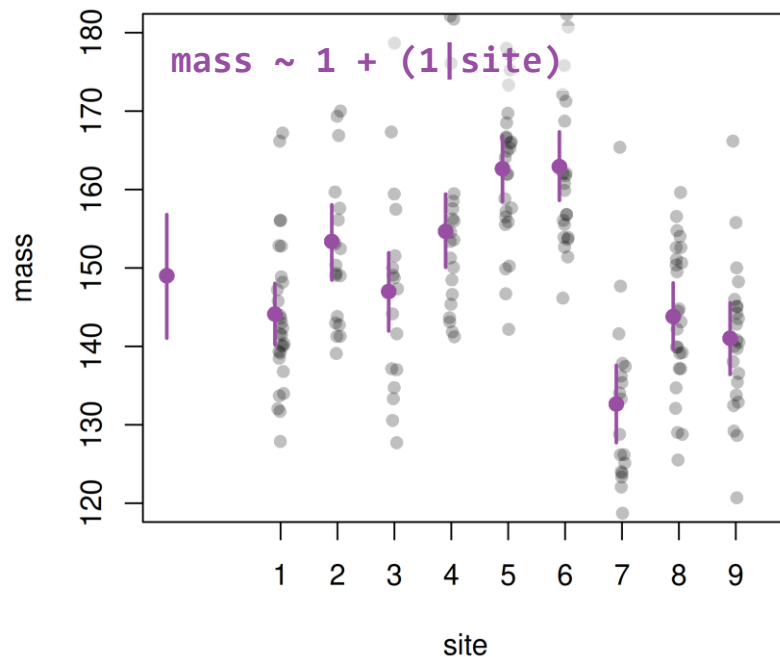
In previous example, we were interested in a correct statistical model for the overall intercept

→ „**Fixed effect**“ μ_a

Variation among sites was not our focus, but we wanted to account for non-independence of observations

→ „**Random effect**“ δ_j ($j = 1 \dots M$)

Fitted an intercept-only model with random intercepts (site)



Example

- Measured weight of $N = 200$ viper snakes
- Data collected on $M = 9$ different sites across France
- **Also measured length (continuous predictor)**

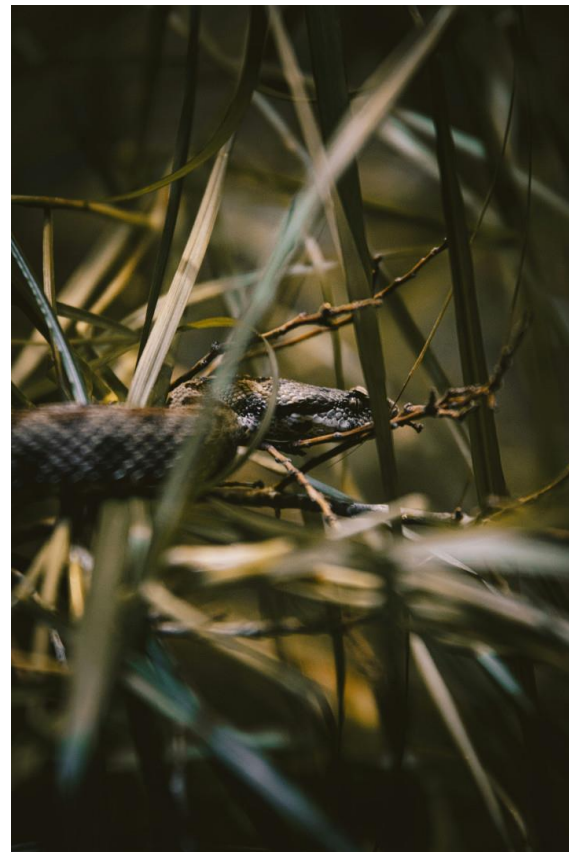
Q: Correct statistical regression model for weight vs. length?

`brm( weight ~ length ) ?`

→ No, observations not independent

`brm( weight ~ length*site ) ?`

→ No, does not estimate overall slope (just site-specific regressions)



(1) Complete pooling

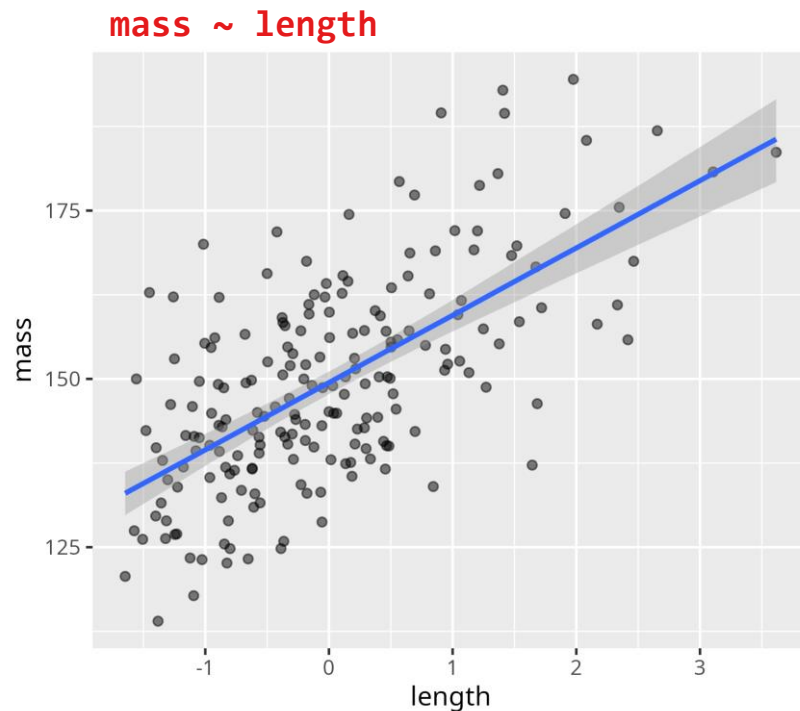
Model: $y_i = a + b \cdot \text{length}_i + \varepsilon_i \quad i = 1 \dots N$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

Priors: $a \sim \text{Normal}(150, 10)$

$b \sim \text{Normal}(10, 5)$

- Ignore categorical predictor *site*
- Information is completely pooled across all levels
- Fit joint intercept a & slope b only
- a, b are given a prior



(2) No pooling

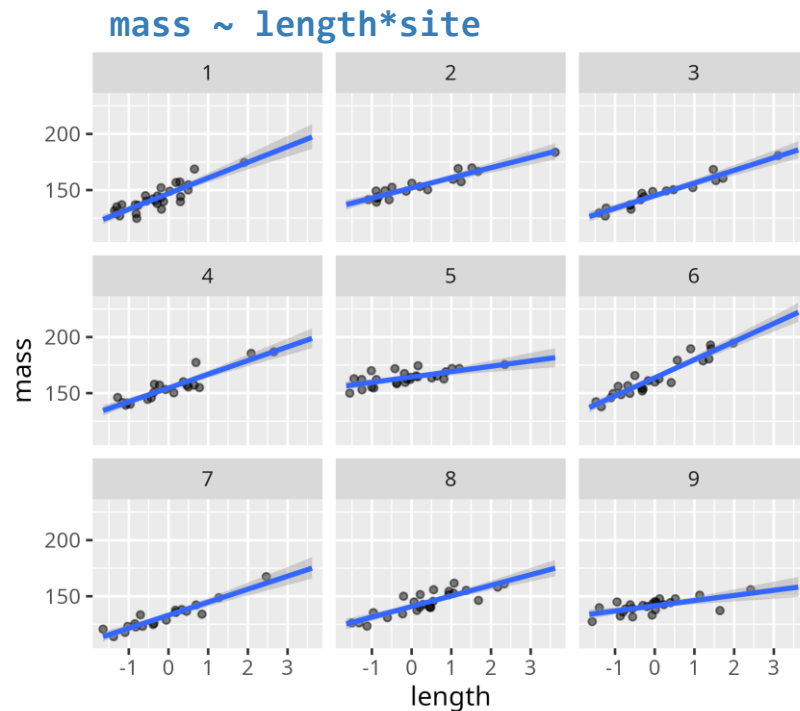
Model: $y_i = a_{site(i)} + b_{site(i)} \cdot length_i + \varepsilon_i$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

Priors: $a_j \sim \text{Normal}(150, 10)$ $j = 1 \dots M$

$b_j \sim \text{Normal}(10, 5)$

- Include categorical predictor *site* (**ANCOVA**)
- No information is shared across levels
- Fit independent intercepts $a_1 \dots a_M$ & slopes $b_1 \dots b_M$
- Each a_j & b_j is given a prior
(here they are identical, but could also be different)



(2) No pooling

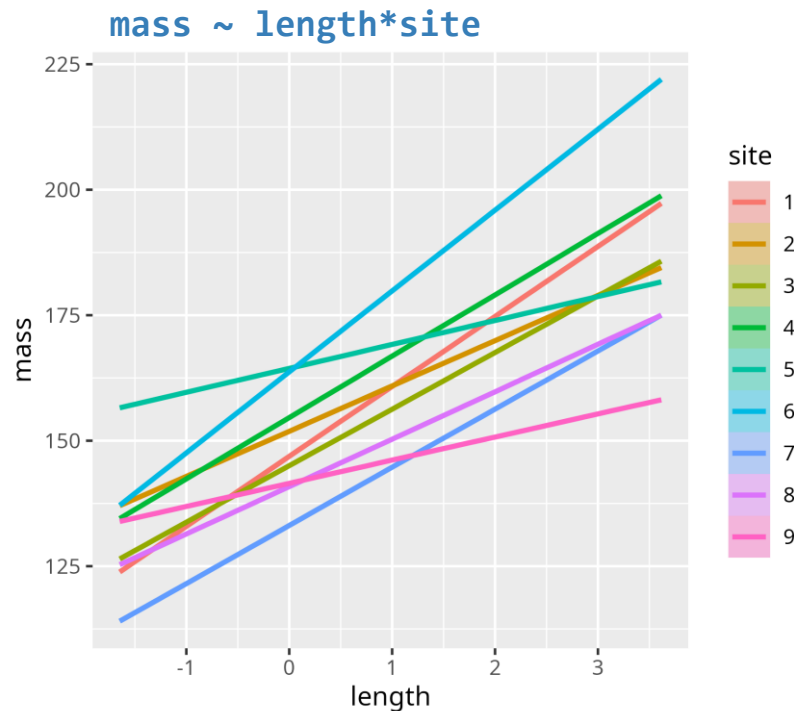
Model: $y_i = a_{site(i)} + b_{site(i)} \cdot length_i + \varepsilon_i$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

Priors: $a_j \sim \text{Normal}(150, 10)$ $j = 1 \dots M$

$b_j \sim \text{Normal}(10, 5)$

- Include categorical predictor *site* (**ANCOVA**)
- No information is shared across levels
- Fit independent intercepts $a_1 \dots a_M$ & slopes $b_1 \dots b_M$
- Each a_j & b_j is given a prior
(here they are identical, but could also be different)



(3) Partial pooling: Random intercepts

Model: $y_i = \mathbf{a}_{\text{site}(i)} + b \cdot \text{length}_i + \varepsilon_i$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

$\mathbf{a}_j \sim \text{Normal}(\mu_a, \sigma_a) \quad j = 1 \dots M$

Priors: $b_j \sim \text{Normal}(10, 5)$

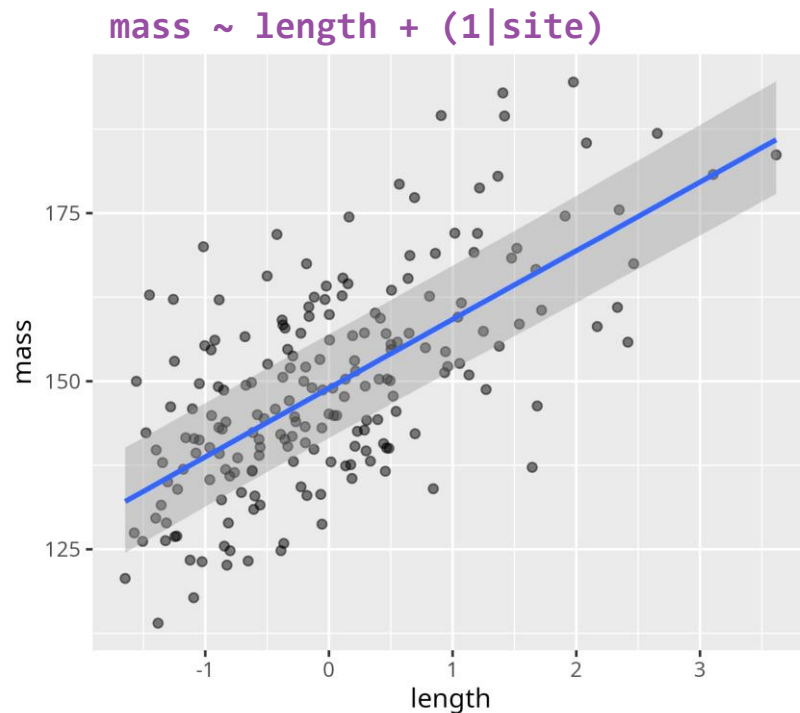
$\mu_a \sim \text{Normal}(150, 10)$

$\sigma_a \sim \text{Exponential}(1/10)$

Random effects: $a_1 \dots a_M$ site-specific intercepts

Fixed effects: μ_a joint intercept

b joint slope



Marginal predictions: $\mu_a + b \cdot \text{length}$

(3) Partial pooling: Random intercepts

Model: $y_i = (\mu_a + \delta_{\text{site}(i)}) + b \cdot \text{length}_i + \varepsilon_i$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

$\delta_j \sim \text{Normal}(0, \sigma_a) \quad j = 1 \dots M$

Priors: $b_j \sim \text{Normal}(10, 5)$

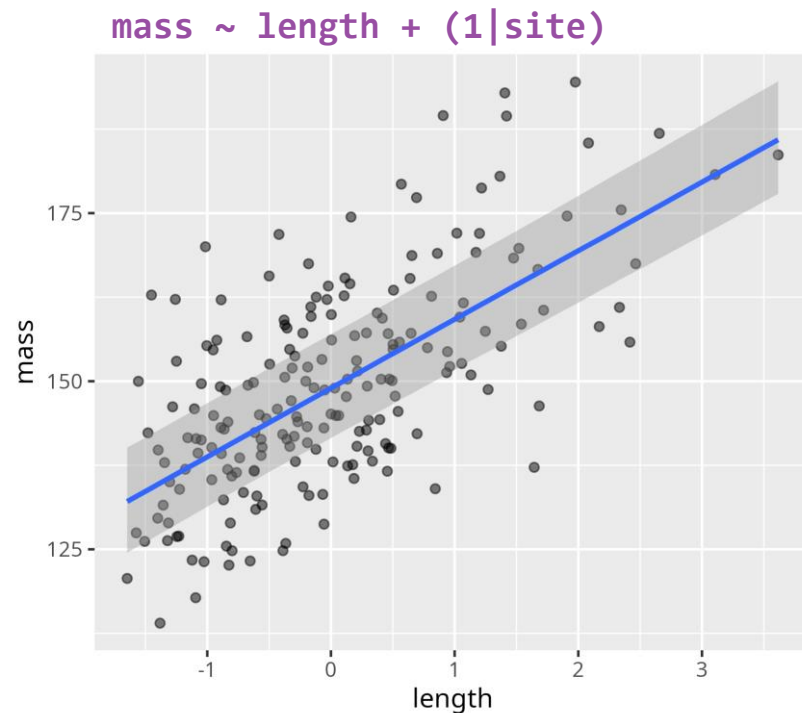
$\mu_a \sim \text{Normal}(150, 10)$

$\sigma_a \sim \text{Exponential}(1/10)$

Random effects: $\delta_1 \dots \delta_M$ intercepts deviation

Fixed effects: μ_a joint intercept

b joint slope



Marginal predictions: $\mu_a + b \cdot \text{length}$

(3) Partial pooling: Random intercepts

Model: $y_i = (\mu_a + \delta_{\text{site}(i)}) + b \cdot \text{length}_i + \varepsilon_i$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

$\delta_j \sim \text{Normal}(0, \sigma_a) \quad j = 1 \dots M$

Priors: $b_j \sim \text{Normal}(10, 5)$

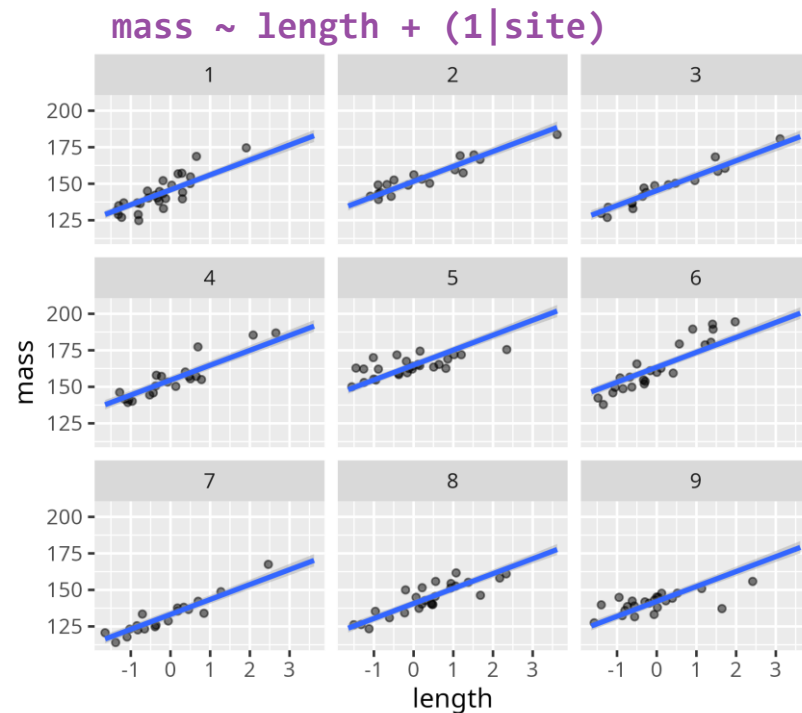
$\mu_a \sim \text{Normal}(150, 10)$

$\sigma_a \sim \text{Exponential}(1/10)$

Random effects: $\delta_1 \dots \delta_M$ intercepts deviation

Fixed effects: μ_a joint intercept

b joint slope



Conditional predictions: $a_j + b \cdot \text{length}$

(3) Partial pooling: Random intercepts

Model: $y_i = (\mu_a + \delta_{\text{site}(i)}) + b \cdot \text{length}_i + \varepsilon_i$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

$\delta_j \sim \text{Normal}(0, \sigma_a) \quad j = 1 \dots M$

Priors: $b_j \sim \text{Normal}(10, 5)$

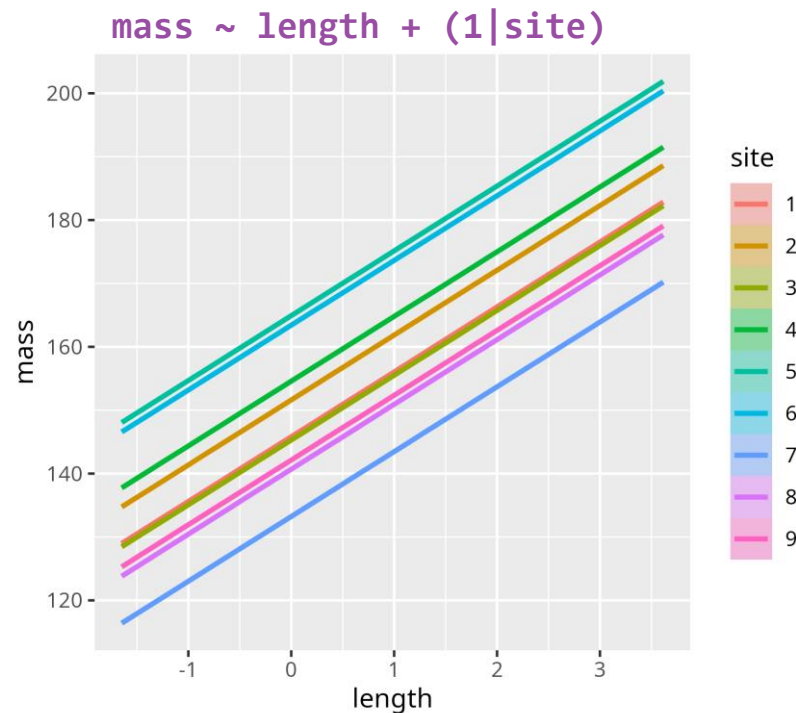
$\mu_a \sim \text{Normal}(150, 10)$

$\sigma_a \sim \text{Exponential}(1/10)$

Random effects: $\delta_1 \dots \delta_M$ intercepts deviation

Fixed effects: μ_a joint intercept

b joint slope



Conditional predictions: $a_j + b \cdot \text{length}$

(4) Partial pooling: Random slopes & intercepts

Model: $y_i = \mathbf{a}_{\text{site}(i)} + \mathbf{b}_{\text{site}(i)} \cdot \text{length}_i + \varepsilon_i$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

$\mathbf{a}_j \sim \text{Normal}(\mu_a, \sigma_a) \quad j = 1 \dots M$

$\mathbf{b}_j \sim \text{Normal}(\mu_b, \sigma_b)$

Priors $\mu_a \sim \text{Normal}(150, 10)$

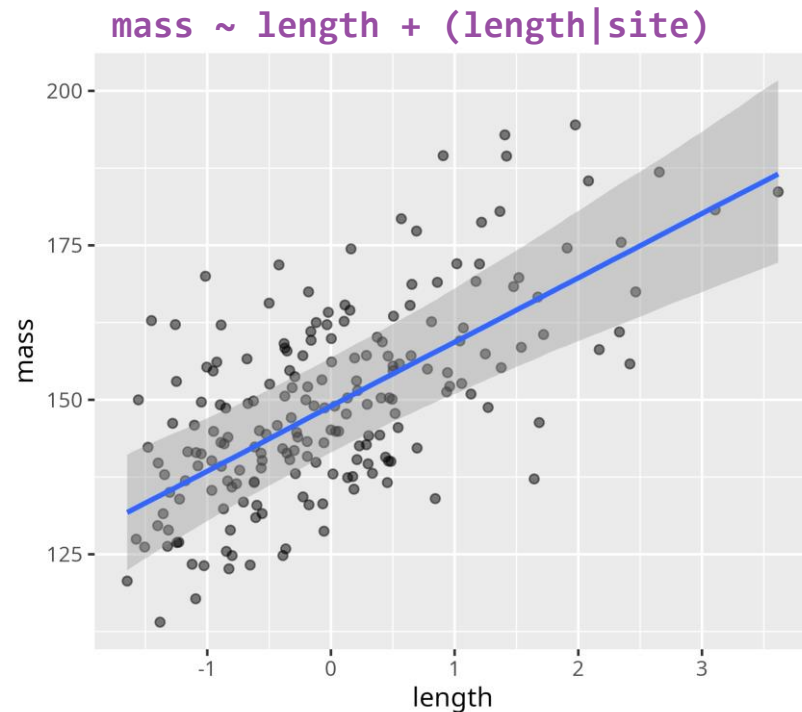
$\mu_b \sim \text{Normal}(10, 5)$

$\sigma_a \sim \text{Exponential}(1/10)$

$\sigma_b \sim \text{Exponential}(1/5)$

Random effects: $a_1 \dots a_M, b_1 \dots b_M$

Fixed effects: μ_a, μ_b joint intercept & slope



Marginal predictions: $\mu_a + \mu_b \cdot \text{length}$

(4) Partial pooling: Random slopes & intercepts

Model: $y_i = (\mu_a + \delta_{site(i)}) + (\mu_b + \gamma_{site(i)}) \cdot length_i + \varepsilon_i$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

$\delta_j \sim \text{Normal}(0, \sigma_a) \quad j = 1 \dots M$

$\gamma_j \sim \text{Normal}(0, \sigma_b)$

Priors $\mu_a \sim \text{Normal}(150, 10)$

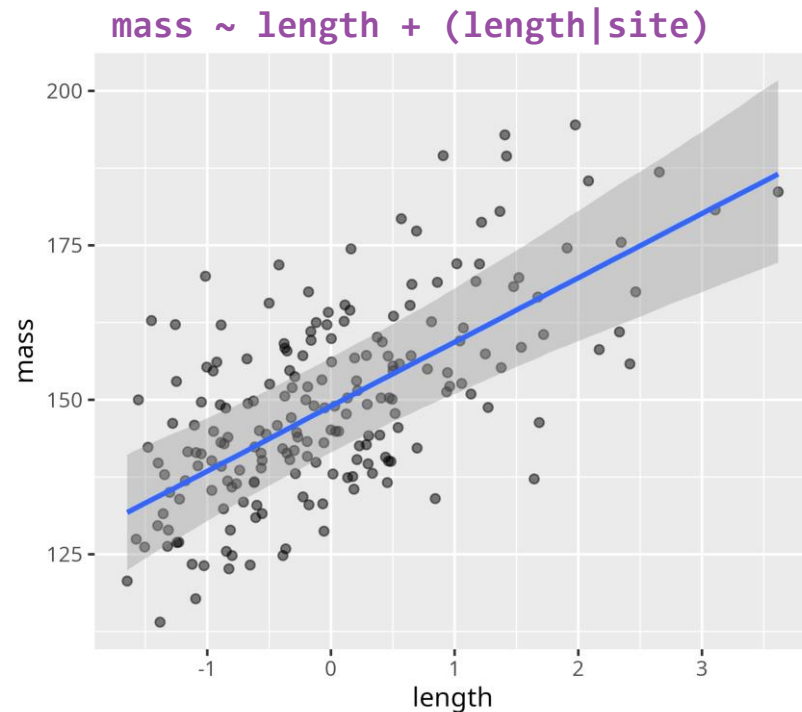
$\mu_b \sim \text{Normal}(10, 5)$

$\sigma_a \sim \text{Exponential}(1/10)$

$\sigma_b \sim \text{Exponential}(1/5)$

Random effects: $\delta_1 \dots \delta_M, \gamma_1 \dots \gamma_M$

Fixed effects: μ_a, μ_b joint intercept & slope



Marginal predictions: $\mu_a + \mu_b \cdot length$

(4) Partial pooling: Random slopes & intercepts

Model: $y_i = (\mu_a + \delta_{site(i)}) + (\mu_b + \gamma_{site(i)}) \cdot length_i + \varepsilon_i$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

$\delta_j \sim \text{Normal}(0, \sigma_a) \quad j = 1 \dots M$

$\gamma_j \sim \text{Normal}(0, \sigma_b)$

Priors $\mu_a \sim \text{Normal}(150, 10)$

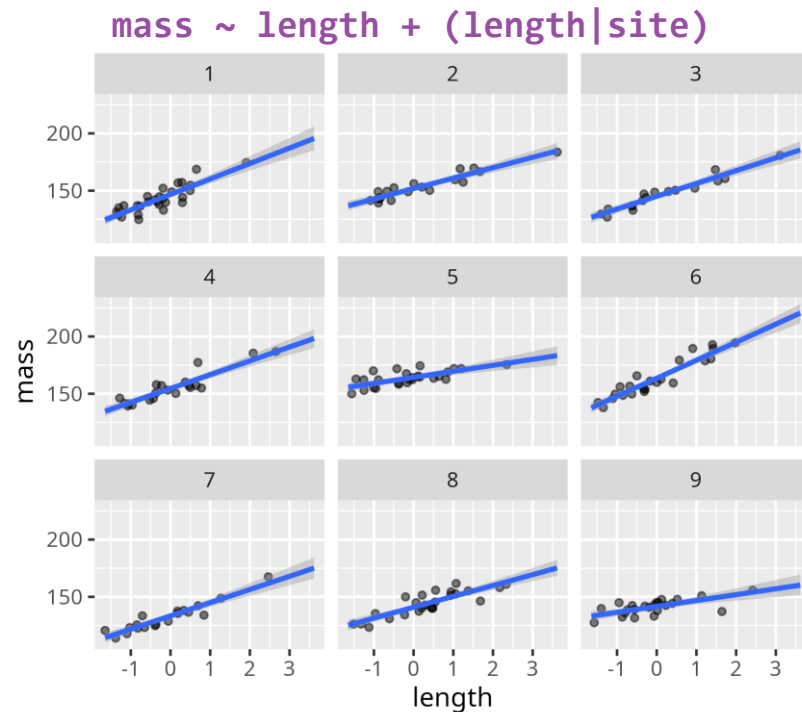
$\mu_b \sim \text{Normal}(10, 5)$

$\sigma_a \sim \text{Exponential}(1/10)$

$\sigma_b \sim \text{Exponential}(1/5)$

Random effects: $\delta_1 \dots \delta_M, \gamma_1 \dots \gamma_M$

Fixed effects: μ_a, μ_b joint intercept & slope



Conditional predictions: $a_j + b_j \cdot length$

(4) Partial pooling: Random slopes & intercepts

Model: $y_i = (\mu_a + \delta_{\text{site}(i)} + (\mu_b + \gamma_{\text{site}(i)}) \cdot \text{length}_i + \varepsilon_i$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

$\delta_j \sim \text{Normal}(0, \sigma_a) \quad j = 1 \dots M$

$\gamma_j \sim \text{Normal}(0, \sigma_b)$

Priors $\mu_a \sim \text{Normal}(150, 10)$

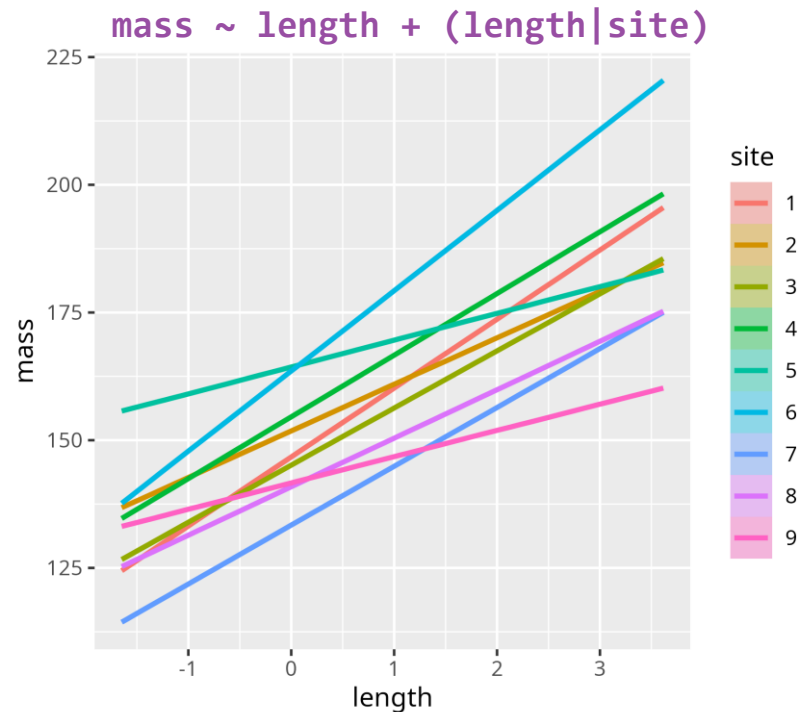
$\mu_b \sim \text{Normal}(10, 5)$

$\sigma_a \sim \text{Exponential}(1/10)$

$\sigma_b \sim \text{Exponential}(1/5)$

Random effects: $\delta_1 \dots \delta_M, \gamma_1 \dots \gamma_M$

Fixed effects: μ_a, μ_b joint intercept & slope



Conditional predictions: $a_j + b_j \cdot \text{length}$

(4) Partial pooling: Random slopes & intercepts

Model: $y_i = (\mu_a + \delta_{site(i)}) + (\mu_b + \gamma_{site(i)}) \cdot length_i + \varepsilon_i$
 $\varepsilon_i \sim \text{Normal}(0, \sigma)$

$$\begin{pmatrix} \delta_j \\ \gamma_j \end{pmatrix} \sim \text{MVNormal} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{bmatrix} \sigma_a^2 & \rho \sigma_a \sigma_b \\ \rho \sigma_a \sigma_b & \sigma_b^2 \end{bmatrix} \right)$$

$j = 1 \dots M$

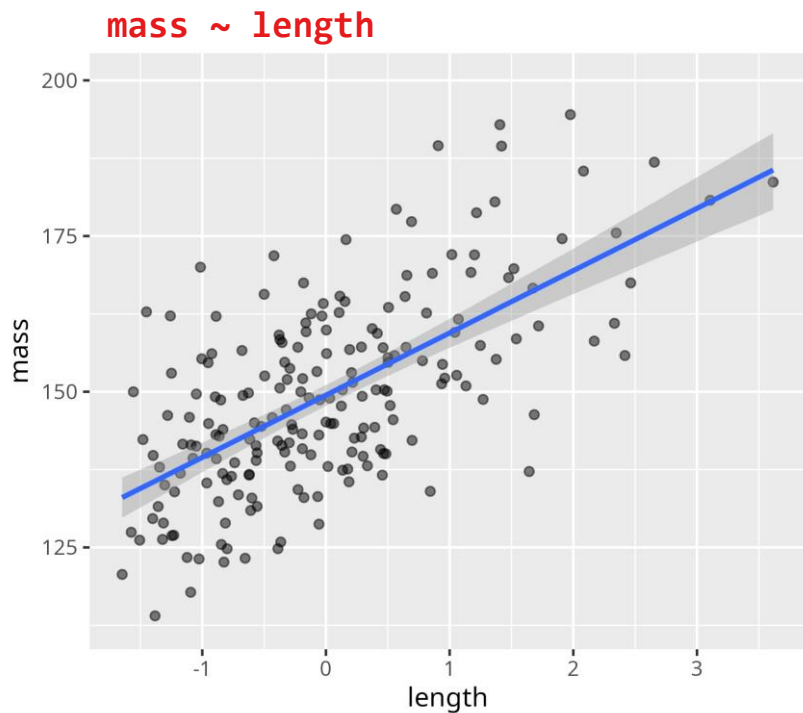
Priors $\mu_a \sim \text{Normal}(150, 10)$
 $\mu_b \sim \text{Normal}(10, 5)$
 $\sigma_a \sim \text{Exponential}(1/10)$
 $\sigma_b \sim \text{Exponential}(1/5)$
 $\rho \sim \text{Uniform}(0, 1)$

← Random intercepts & slopes not independent:

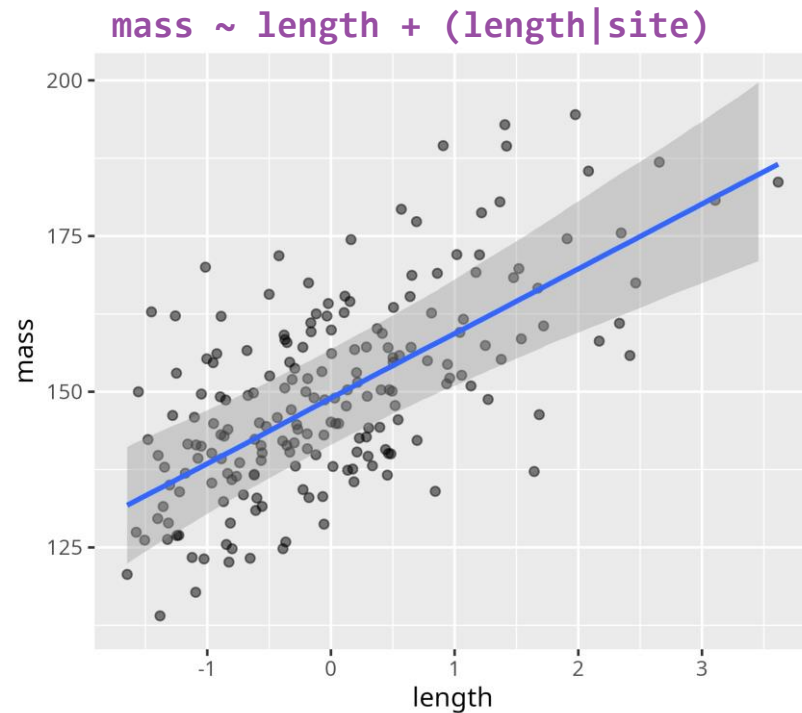
Multivariate normal distributions with zero mean and 2x2 covariance matrix Σ

← Correlation coefficient ρ

Comparison



Uncertainty of a, b is smaller (overly confident).
Wrongly assumes $N = 200$ independent obs.

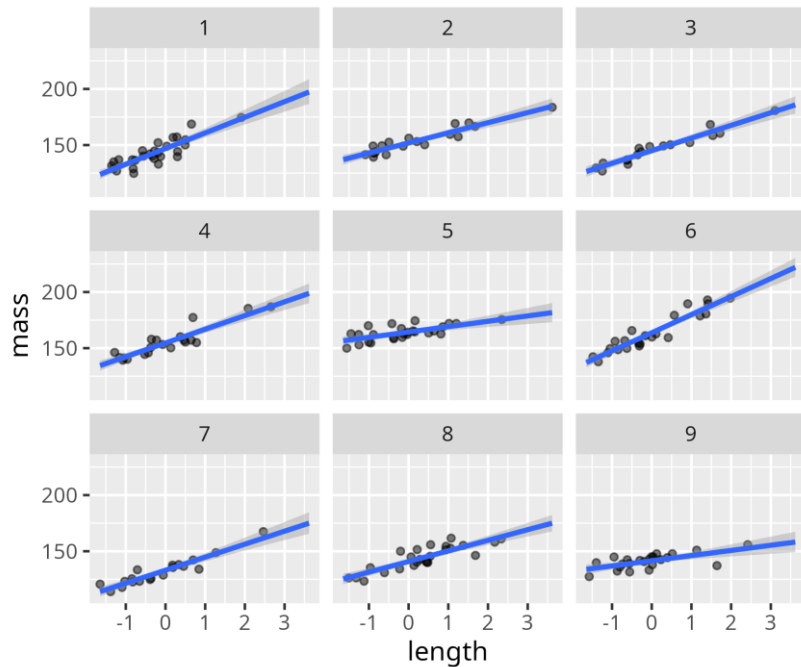


Uncertainty of μ_a, μ_b is correct.

Comparison

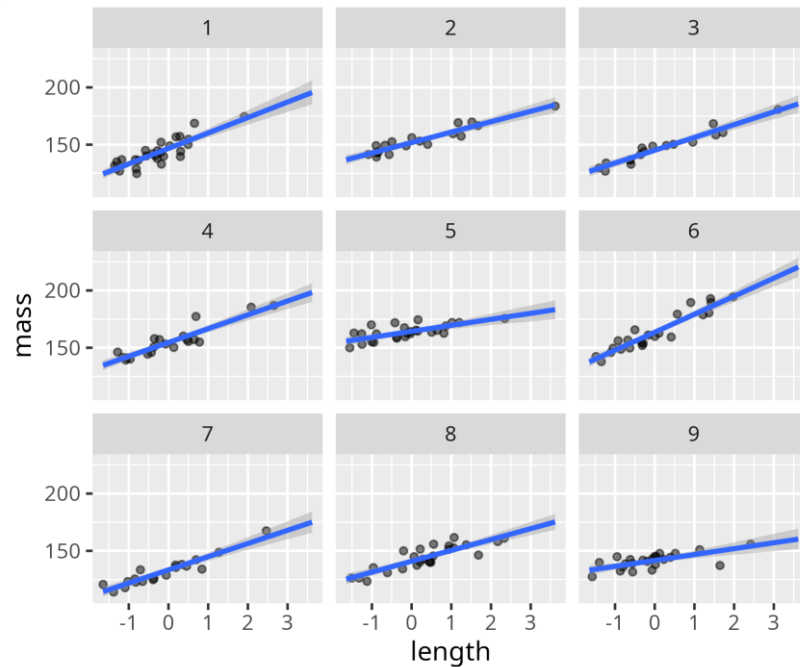
Here,
difference is
very weak

$\text{mass} \sim \text{length} * \text{site}$



Regressions for each site independent.

$\text{mass} \sim \text{length} + (\text{length} | \text{site})$

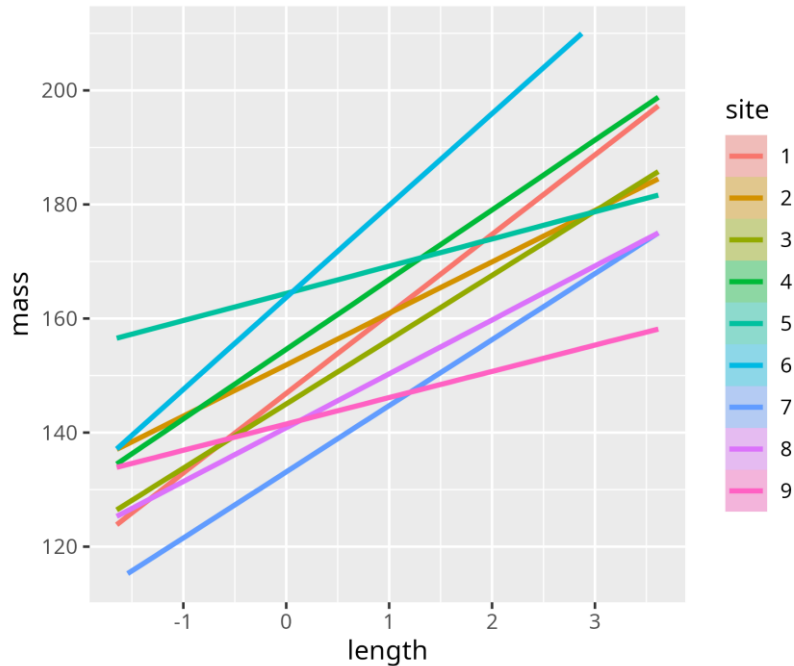


Intercepts & slopes draws to joint means
„Shrinkage“

Comparison

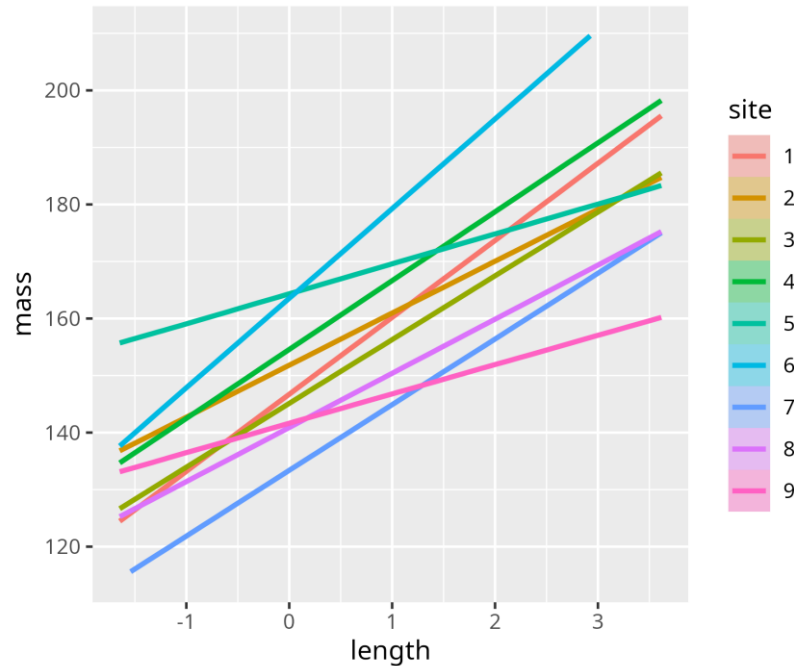
Here,
difference is
very weak

$\text{mass} \sim \text{length} * \text{site}$



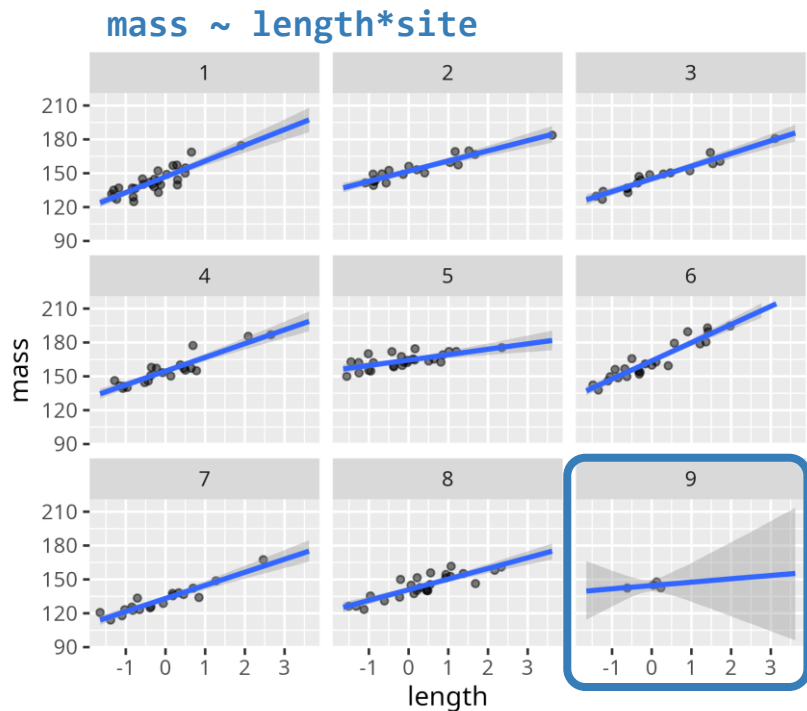
Regressions for each site independent.

$\text{mass} \sim \text{length} + (\text{length} | \text{site})$

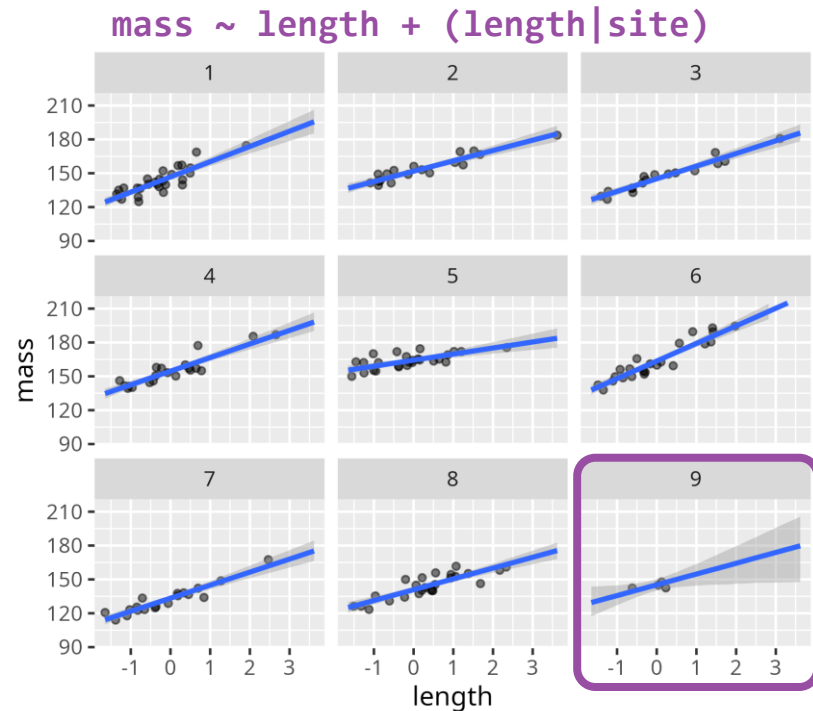


Intercepts & slopes draws to joint means
„Shrinkage“

Comparison: unbalanced design



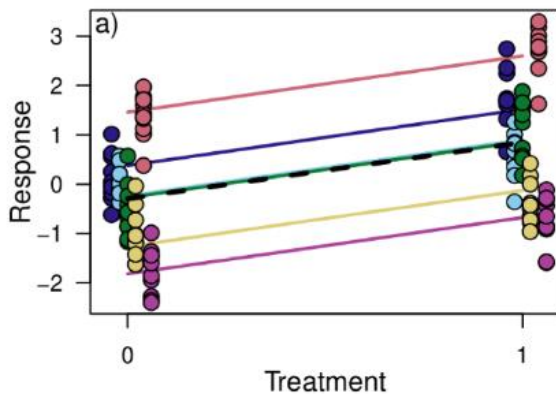
High uncertainty in sites with few data.



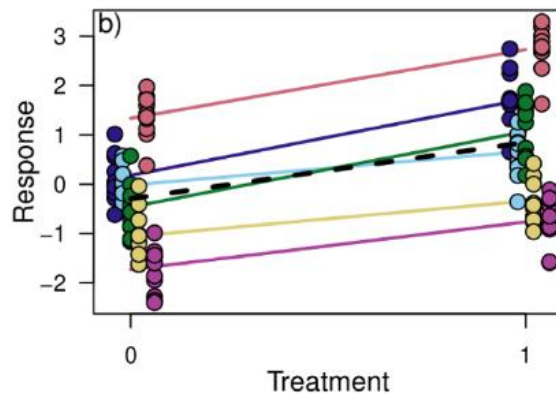
Borrowing strength from other sites.
→ Better predictions!

**Categorical
predictor**

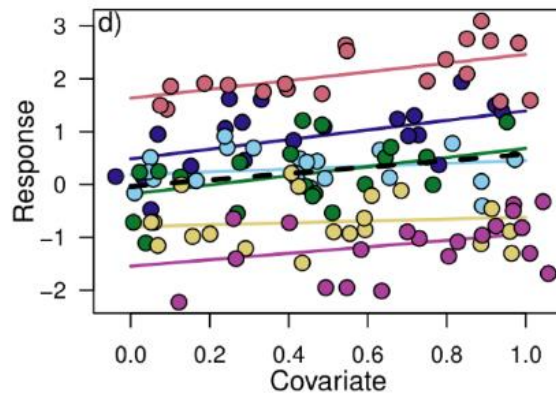
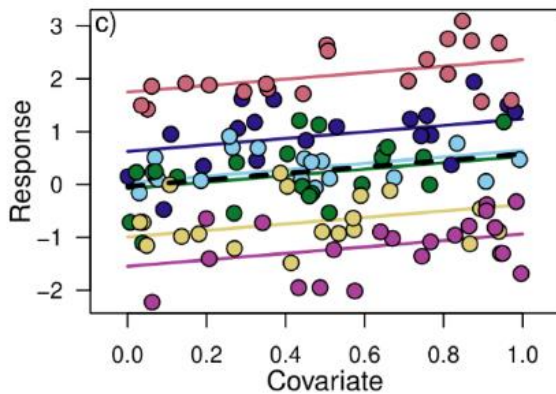
Random intercepts



Random intercepts & „slopes“



**Continuous
predictor**



Harrison et al
(2020) *PeerJ*

```
> brm(mass ~ length + (length|site), ...)
```

```
Family: gaussian
Links: mu = identity; sigma = identity
Formula: mass ~ length + (length | site)
Data: df (Number of observations: 200)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

~site (Number of levels: 9)

		Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
$\sigma_a \rightarrow$	sd(Intercept)	12.36	3.45	7.39	20.97	1.00	1269
$\sigma_b \rightarrow$	sd(length)	4.65	1.57	2.52	8.72	1.00	1154
$\rho \rightarrow$	cor(Intercept,length)	0.05	0.33	-0.58	0.64	1.00	1937

Regression Coefficients:

		Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
$\mu_a \rightarrow$	Intercept	148.95	3.88	141.48	156.81	1.00	1265	2055
$\mu_b \rightarrow$	length	10.43	1.69	7.11	13.81	1.00	1318	1461

Further Distributional Parameters:

		Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
$\sigma \rightarrow$	sigma	5.28	0.27	4.79	5.86	1.00	3411	2993

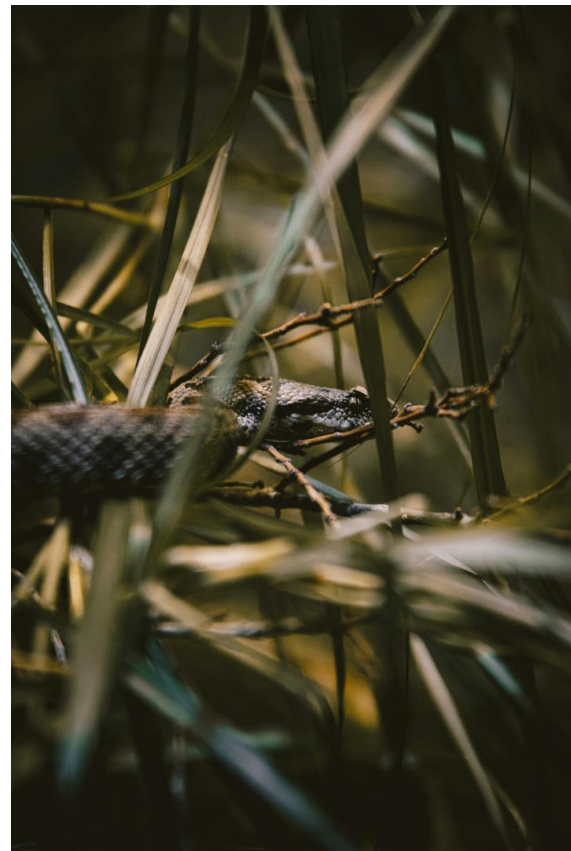
Multiple grouping factors

Example

- Measured weight of $N = 200$ viper snakes
- Data collected on $M = 9$ different sites across France
- **Collection was done in $L = 5$ sampling campaigns**

Can be fully crossed (each campaign visited all sites)
or partially crossed (not all campaigns visited all sites).

→ Specify **additive (independent)** random effects
for grouping factor site and grouping factor campaign



Crossed random effects

Model: $y_i = \mu_a + \delta_{site(i)} + \gamma_{campaign(i)} + \varepsilon_i$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

$\delta_j \sim \text{Normal}(0, \sigma_\delta) \quad j = 1 \dots M$

$\gamma_k \sim \text{Normal}(0, \sigma_\gamma) \quad k = 1 \dots L$

Priors: $\mu_a \sim \text{Normal}(150, 10)$

$\sigma_\delta \sim \text{Exponential}(1/10)$

$\sigma_\gamma \sim \text{Exponential}(1/10)$

$\text{mass} \sim 1 + (1|\text{site}) + (1|\text{campaign})$

← Site-effect

← Campaign-effect

← Overall mean is the only fixed effect
in this basic example

Variance partitioning

Overall variance of the data is estimated on different levels

$$s_{\text{total}}^2 = s_{\text{among}}^2 + s_{\text{within}}^2$$

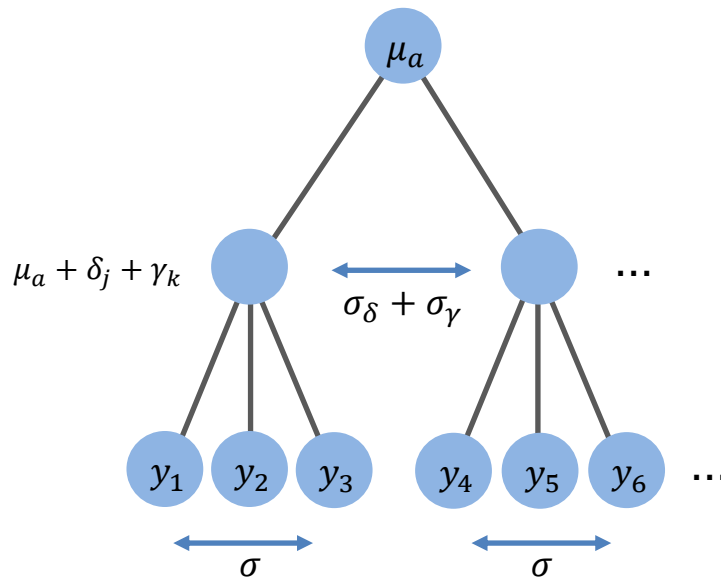
Site-campaign-level (among) $\sigma_{\delta} + \sigma_{\gamma}$

Explains additive deviations from overall mean μ_a
caused by site (σ_{δ}) and campaign (σ_{γ})

Observation-level (within) σ

Explains each data point's deviation from
site-campaign-mean $\mu_a + \delta_j + \gamma_k$

(All the randomness in the data which is not caused by site-level differences)



> `brm(mass ~ 1 + (1|site) + (1|campaign), ...)`

Family: gaussian

Links: mu = identity; sigma = identity

Formula: `mass ~ 1 + (1 | site) + (1 | campaign)`

Data: df (Number of observations: 200)

Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;

total post-warmup draws = 4000

Multilevel Hyperparameters:

~campaign (Number of levels: 5)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
$\sigma_\gamma \rightarrow$ sd(Intercept)	1.39	1.33	0.05	4.94	1.00	1983	2233

~site (Number of levels: 9)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
$\sigma_\delta \rightarrow$ sd(Intercept)	11.38	3.22	6.89	19.27	1.01	1085	2004

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
$\mu_a \rightarrow$ Intercept	149.14	3.86	141.37	156.68	1.00	828	1274

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
$\sigma \rightarrow$ sigma	11.04	0.58	9.99	12.20	1.00	3320	2678

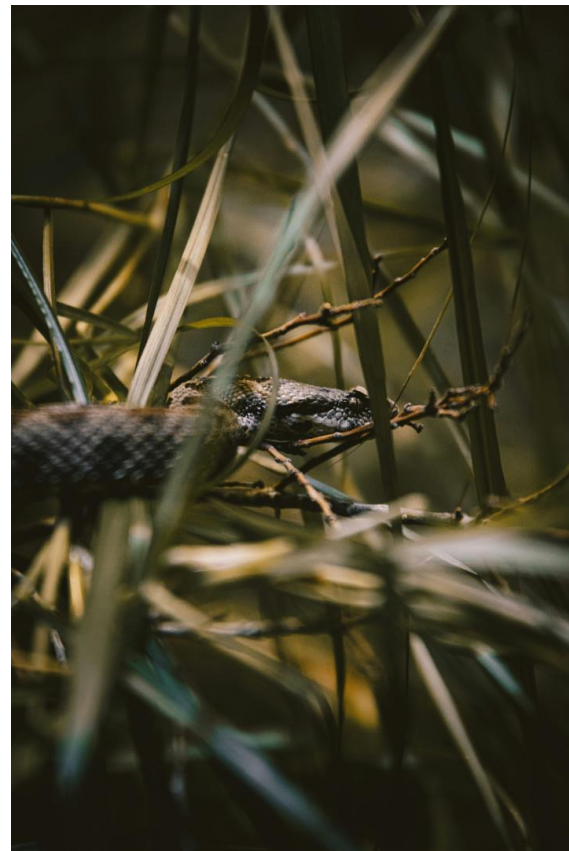
Example

- Measured weight of $N = 200$ viper snakes
- Data collected on $M = 9$ different sites across France
- **Sites belong to $L = 3$ distinct mountain regions**

Every site uniquely belongs to one region.

At least one region has 2 or more sites.

→ Specify **nested** random effects
for factor site in factor region



Nested random effects

Model: $y_i = \mu_a + \gamma_{\text{region}(i)} + \delta_{\text{region:site}(i)} + \varepsilon_i$

$\varepsilon_i \sim \text{Normal}(0, \sigma)$

$\gamma_k \sim \text{Normal}(0, \sigma_k) \quad k = 1 \dots L$

$\delta_j \sim \text{Normal}(0, \sigma_\delta) \quad j = 1 \dots M$

Priors: $\mu_a \sim \text{Normal}(150, 10)$

$\sigma_\gamma \sim \text{Exponential}(1/10)$

$\sigma_\delta \sim \text{Exponential}(1/10)$

$\text{mass} \sim 1 + (1|\text{region/site})$

$\text{mass} \sim 1 + (1|\text{region}) + (1|\text{region:site})$

← Regional-effect

← Site-effect in region

Automatically
uses unique
region:site
combinations
with M levels

Variance partitioning

Overall variance of the data is estimated on different levels

$$s_{\text{total}}^2 = s_{\text{among regions}}^2 + s_{\text{among sites in region}}^2 + s_{\text{within}}^2$$

Regional-level σ_γ

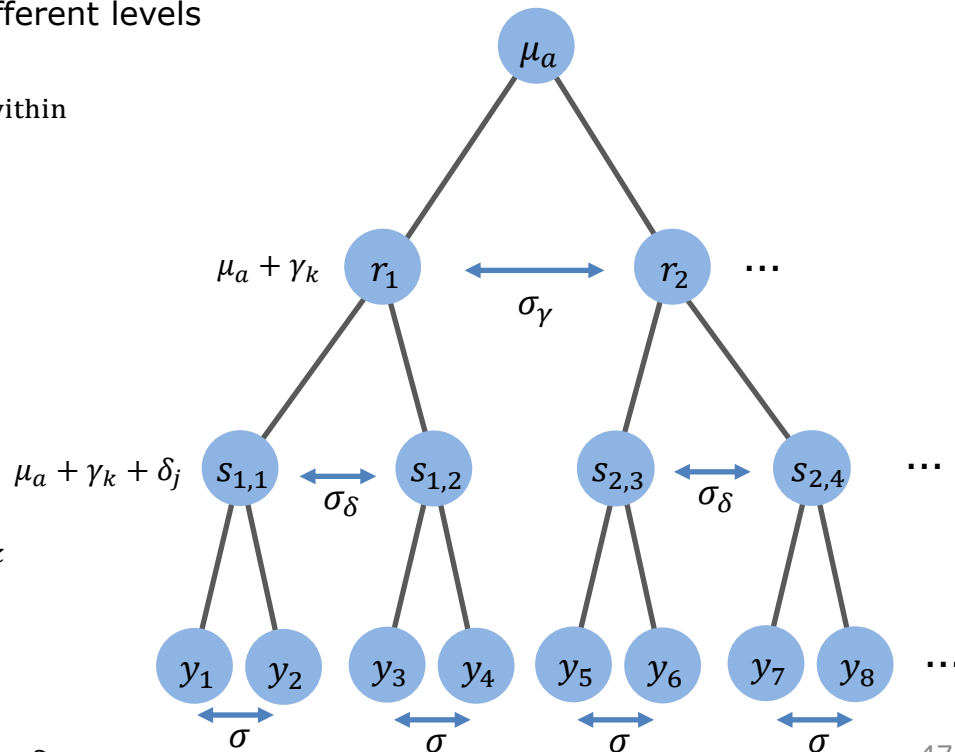
Explains deviations from overall mean μ_a

Site-level σ_δ

Explains deviations from regional mean $\mu_a + \gamma_k$

Observation-level (within) σ

Data point deviations from site-mean $\mu_a + \gamma_k + \delta_j$



```
> brm(mass ~ 1 + (1|region/site), ...)
> brm(mass ~ 1 + (1|region) + (1|region:site), ...)
```

```
Family: gaussian
Links: mu = identity; sigma = identity
Formula: mass ~ 1 + (1 | region/site)
Data: df (Number of observations: 200)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

~region (Number of levels: 3)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
$\sigma_\gamma \rightarrow$ sd(Intercept)	12.21	6.76	2.81	30.38	1.00	1155	1073

~region:site (Number of levels: 9)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
$\sigma_\delta \rightarrow$ sd(Intercept)	6.01	2.60	2.57	12.65	1.00	1003	1395

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
$\mu_a \rightarrow$ Intercept	148.99	6.45	135.11	161.86	1.00	1594	1365

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
$\sigma \rightarrow$ sigma	11.05	0.57	10.01	12.21	1.00	3750	2543

Nested or crossed?

Nested designs

$(1|F1/F2)$ just short for $(1|F1)+(1|F1:F2)$

If nested factor **F2** is uniquely labelled

(levels of **F2** and **F1:F2** identical)

this is same as crossed $(1|F1)+(1|F2)$

Crossed designs

No need to be fully crossed for additive model.

$(1|F1)+(1|F2)$ works for partially crossed, too.

Table 1. Schematic illustration of crossed and nested designs

Nested design $(1 F1/F2)$				
Factor 2				
Factor 1	a	b	c	d
A	X	X		
B			X	X

Partially crossed design $(1 F1)+(1 F2)$				
Factor 2				
Factor 1	a	b	c	d
A	X	X		
B	X	X		
C			X	X
D			X	X

Fully crossed design $(1 F1)+(1 F2)$				
Factor 2				
Factor 1	a	b	c	d
A	X	X	X	X
B	X	X	X	X
C	X	X	X	X
D	X	X	X	X

Continuous correlation structures

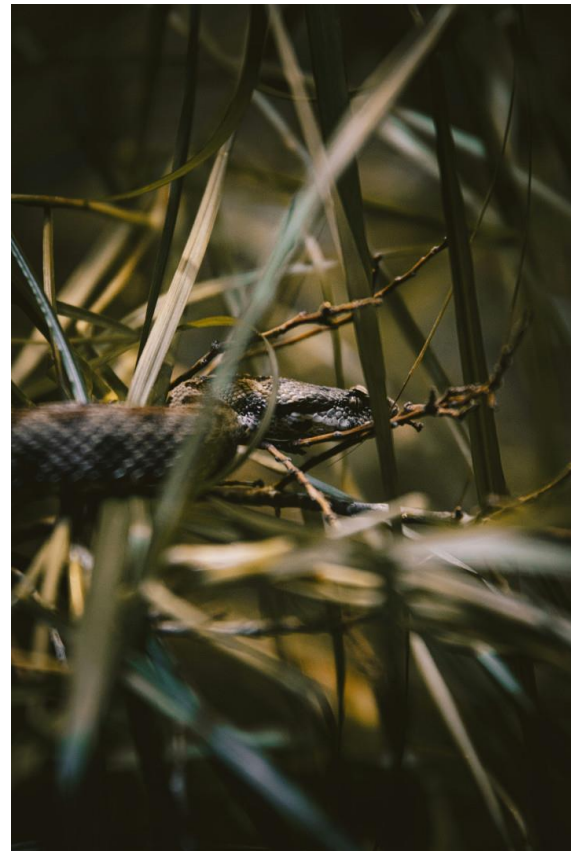
Example

- Measured weight of $N = 200$ viper snakes
- Data collected on $M = 10$ consecutive years (same region)

Instead of using year as categorical grouping predictor, we could use it as continuous „grouping“ predictor.

„Everything is related to everything else,
but near things are more related than distant things.“ (Tobler)

→ Specify temporal correlation model.



Old: Year categorical

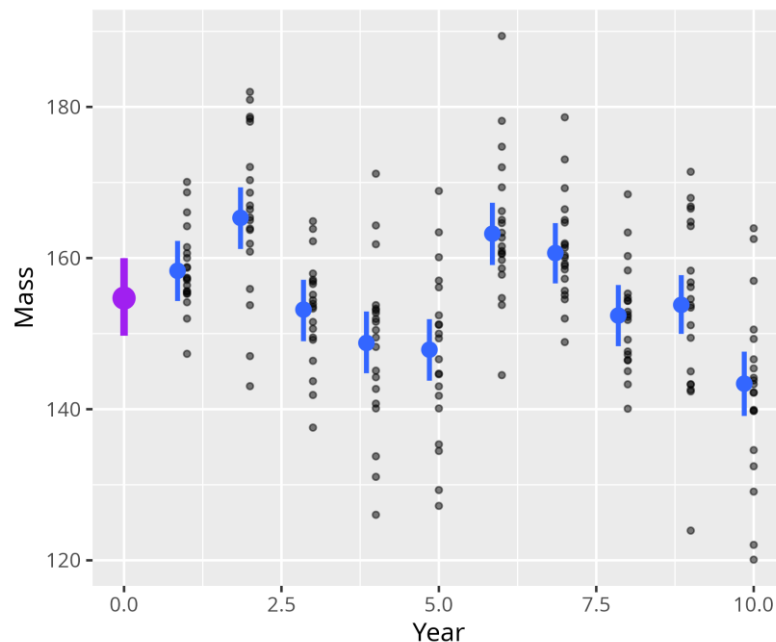
Model: $y_i = \mu + \delta_{\text{year}(i)} + \varepsilon_i \quad i = 1 \dots N$



Random: $\delta_j \sim \text{Normal}(0, \sigma_\delta) \quad j = 1 \dots M$

Priors: $\mu, \sigma, \sigma_\delta$

$\text{mass} \sim 1 + (1|\text{year})$



Does account for grouping of data,
but not for temporal autocorrelation !

New: Year continuous with Gaussian process

Model:

$$y_i = \mu + gp(year_i) + \varepsilon_i$$



Fixed effect



Random effect



Residual

Random:

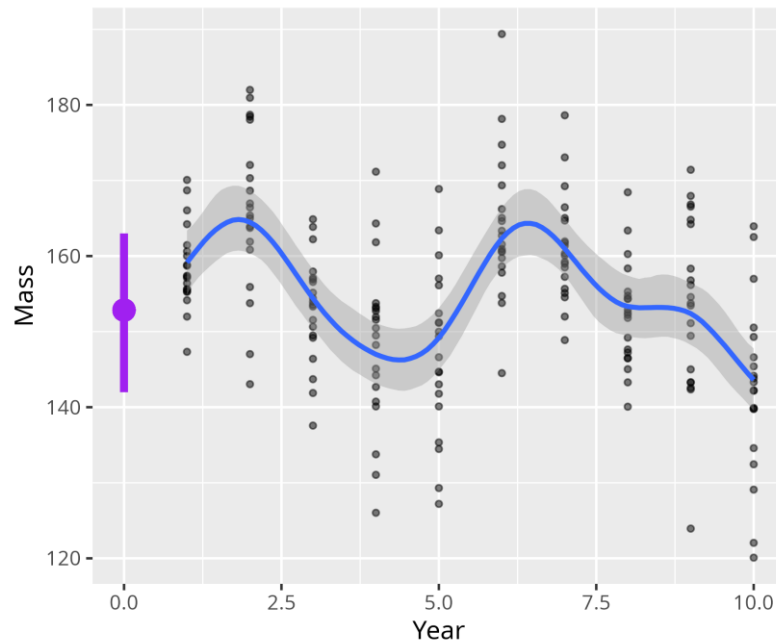
$$E[gp] = 0, \text{Cov}[gp] = \sigma_{gp} \cdot R$$

$$R_{ij} = \exp\left(-\frac{\text{dist}(i,j)^2}{\theta^2}\right)$$

Priors:

$$\mu, \sigma, \sigma_\delta, \theta$$

mass ~ 1 + gp(year)



New: Year continuous with Gaussian process

Model:

$$y_i = \mu + gp(year_i) + \varepsilon_i$$



Fixed effect



Random effect



Residual

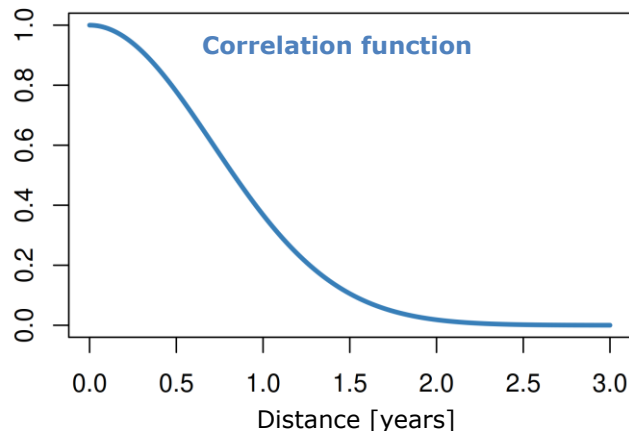
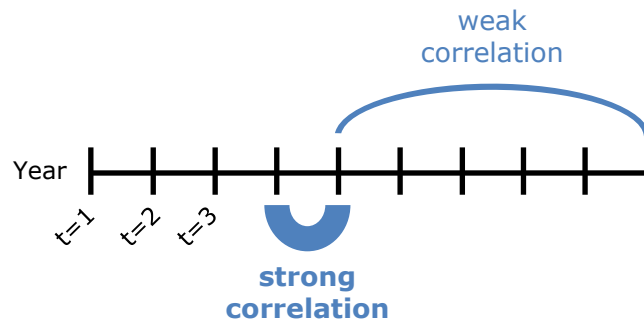
Random:

$$E[gp] = 0, \text{Cov}[gp] = \sigma_{gp} \cdot R$$

$$R_{ij} = \exp\left(-\frac{\text{dist}(i,j)^2}{\theta^2}\right)$$

Priors:

$$\mu, \sigma, \sigma_{gp}, \theta$$



Truly continuous data: Date

Model:

$$y_i = \mu + gp(date_i) + \varepsilon_i$$



Fixed effect



Random effect



Residual

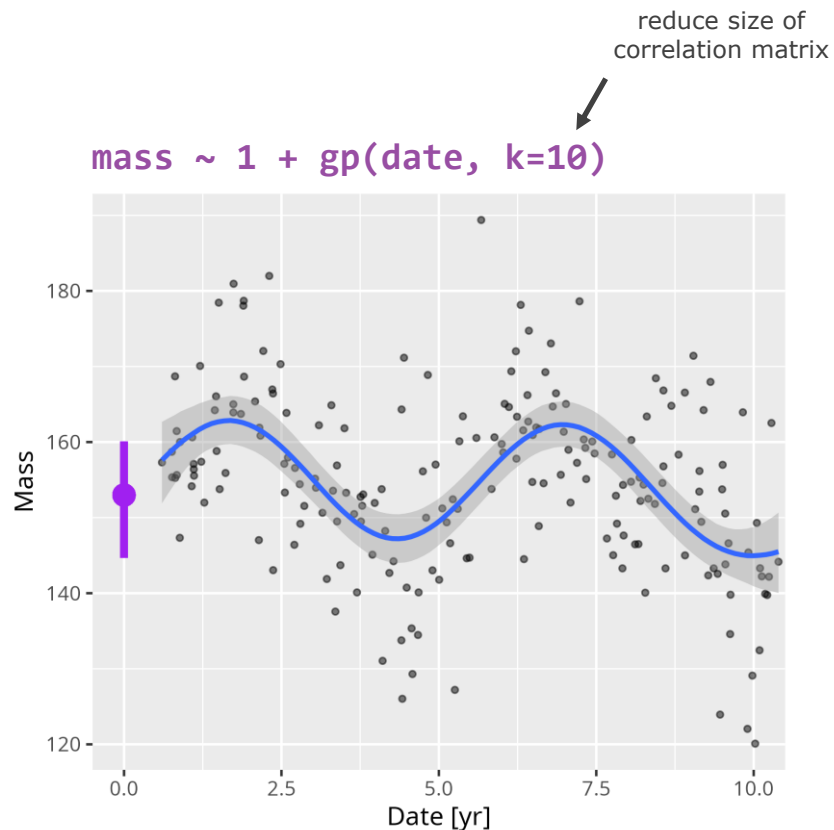
Random:

$$E[gp] = 0, \text{Cov}[gp] = \sigma_{gp} \cdot R$$

$$R_{ij} = \exp\left(-\frac{\text{dist}(i,j)^2}{\theta^2}\right)$$

Priors:

$$\mu, \sigma, \sigma_\delta, \theta$$



Spatial correlation

Model: $y_i = \mu + gp(lon_i, lat_i) + \varepsilon_i$

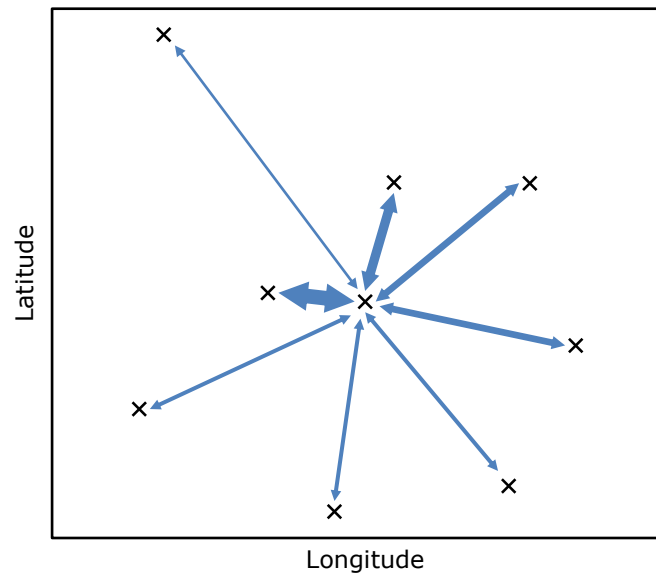
↓ ↓ ↓

Fixed effect Random effect Residual

Random: $E[gp] = 0, Cov[gp] = \sigma_{gp} \cdot R$

$$R_{ij} = \exp\left(-\frac{dist(i,j)^2}{\theta^2}\right)$$

Priors: $\mu, \sigma, \sigma_\delta, \theta$



Spatial correlation

Model: $y_i = \mu + \text{gp}(\text{lon}_i, \text{lat}_i) + \varepsilon_i$

↓ Fixed effect

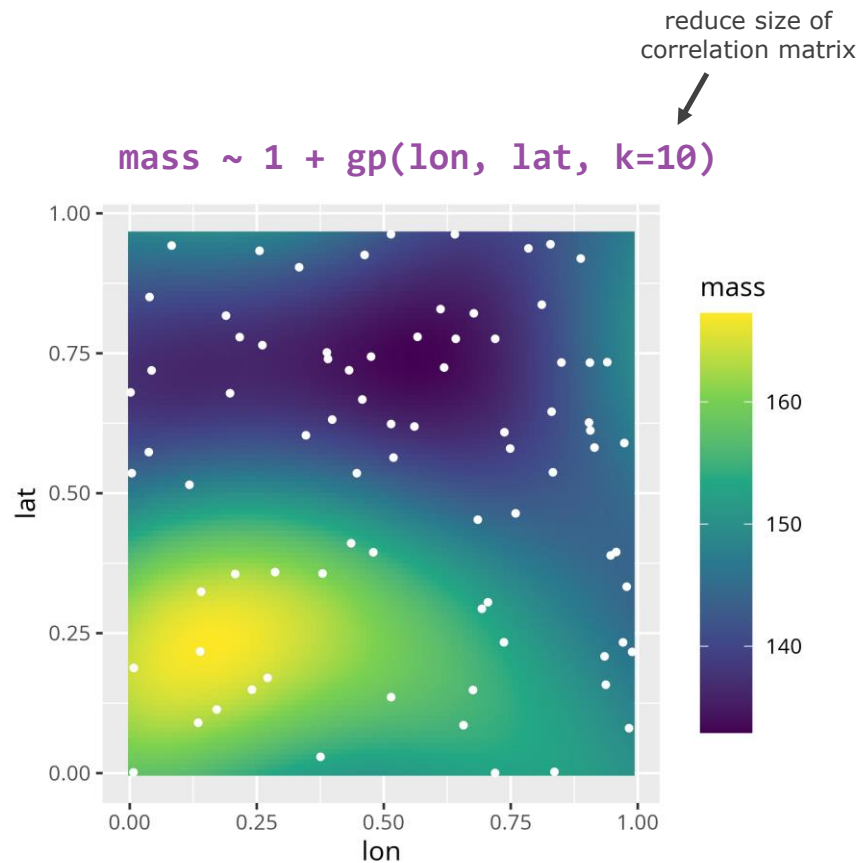
↓ Random effect

↓ Residual

Random: $E[\text{gp}] = 0, \text{Cov}[\text{gp}] = \sigma_{\text{gp}} \cdot R$

$$R_{ij} = \exp\left(-\frac{\text{dist}(i,j)^2}{\theta^2}\right)$$

Priors: $\mu, \sigma, \sigma_{\delta}, \theta$

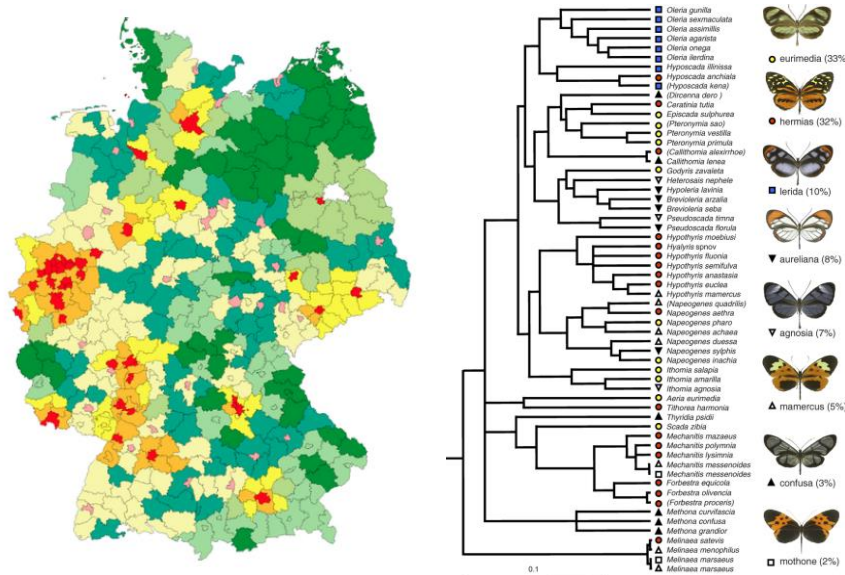


More covariance structures

- Fixed correlation model
- Generalized additive models (GAM)
→ but see mgcv package
- Autoregressive models (AR / ARMA)
for time series
- Conditional autoregressive model (CAR):

$$\text{Adjacency matrix } R_{ij} = \begin{cases} 1 & i, j \text{ neighbours} \\ 0 & \text{otherwise} \end{cases}$$

- Mix & match correlation structures:
spatial, temporal, spatio-temporal,
phylogenetic



Many of these models can get **very slow** even for moderate N .

→ **INLA** is designed for that. Very efficient !

Summary

Random or fixed effects ?

Classical / oldschool interpretation

You are interested in overall means / overall effects, but must account for non-independence in data?

→ **site as a random grouping factor**

~ 1 + (1|site) random intercepts
~ x + (1|site) random intercepts
~ x + (x|site) random intercepts & slopes

You are interested in differences in means / differences in effects between sites?

→ **site as fixed effect**

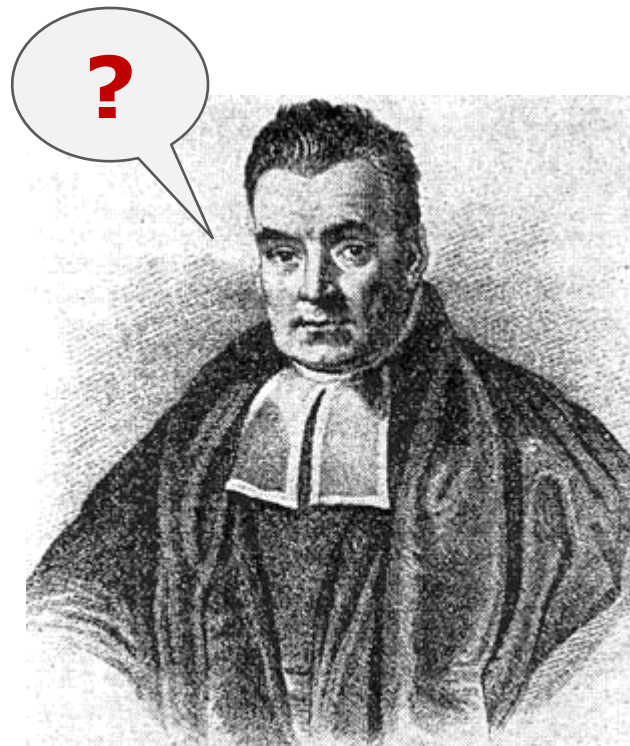
~ site ANOVA
~ x + site ANCOVA (diff. in intercepts)
~ x * site ANCOVA (diff. in intercepts & slopes)

Modern / relaxed interpretation

Can use random factor models for both to overcome difficulties like unbalanced / heterogeneous data

What's Bayesian about it ?

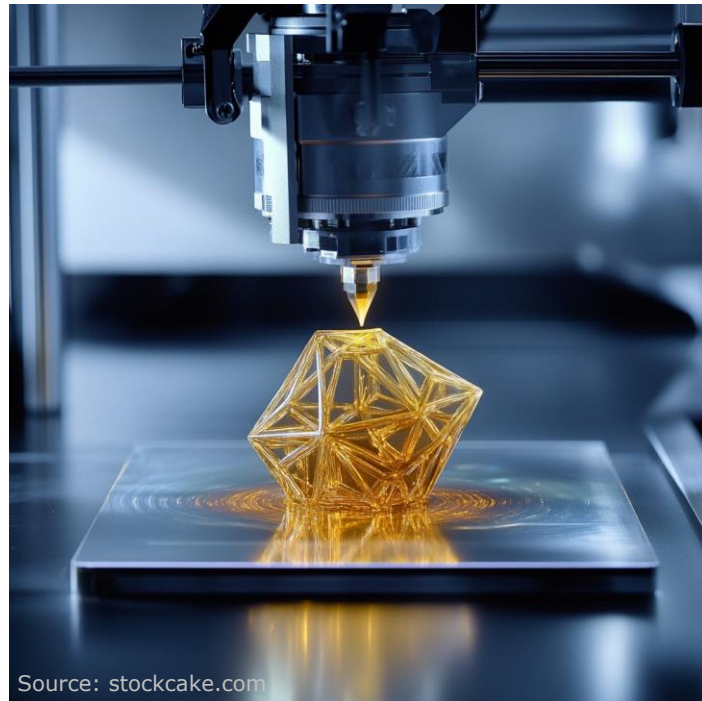
- Sometimes lme4 just does not converge !
- Correct quantification of uncertainties *on all levels*.
- Works for all numbers of groups or numbers of obs.
Already 2-level grouping factor is OK.
- Interpretation “easy” through levels of priors
- Frequentist assumption: Random effects drawn from a larger population of other, unmeasured groups
- Bayes: no such limitation
- Non-normal distribution of random effects?
→ Bayes (Stan) can handle that!



Summary

- Bayesian stats does not free you from model assumptions !
- If residuals are not independent, you must account for it.
- Otherwise: Fixed effects wrongly confident, CIs too narrow
- Thinking hard about your data will reveal random effects structure.
- Formula notation just as in lme4.
- Combine any kind of (generalized) linear (or nonlinear) model with discrete (random) grouping factors and/or continuous correlation structure ...
- ... in one R-package !!

The Bayesian 3D printer



Source: stockcake.com

Further reading

Bolker, B. M. (2015). Linear and generalized linear mixed models. In Fox, G. A. , Negrete-Yankelevich, S. and Sosa, V. J. (eds). *Ecological Statistics: Contemporary theory and application*. *Oxford Academic*. <https://doi.org/10.1093/acprof:oso/9780199672547.003.0014>

Fieberg, J. (2024). *Statistics 4 Ecologists*. <https://statistics4ecologists-v2.netlify.app/> [Chapter 18]

Gelman, A., Hill, J. (2006). *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Cambridge University Press
<http://www.stat.columbia.edu/~gelman/arm/>

Harrison, X. A., Donaldson, L., Correa-Cano, M. E., Evans, J., Fisher, D. N., Goodwin, C. E. D., Robinson, B. S., Hodgson, D. J., & Inger, R. (2018). A brief introduction to mixed effects modelling and multi-model inference in ecology. *PeerJ*, 6, e4794.
<https://doi.org/10.7717/peerj.4794>

Inchausti, P. (2023). *Statistical Modeling With R: a dual frequentist and Bayesian approach for life scientists*. *Oxford University Press*. [Chapter 13]

Kery, M. & Kellner, F. (2024): *Applied Statistical Modelling for Ecologists*. *Elsevier*. [Chapter 10]

Marshall, D. J. (2024). Principles of experimental design for ecology and evolution. *Ecology Letters*, 27(4), e14400.
<https://doi.org/10.1111/ele.14400>

McElreath, R. (2020). *Statistical Rethinking: A Bayesian Course with Examples in R and STAN* (2nd ed.). *Chapman and Hall/CRC*.
<https://doi.org/10.1201/9780429029608> [Chapter 13]

Silk, M. J., Harrison, X. A., & Hodgson, D. J. (2020). Perils and pitfalls of mixed-effects regression models in biology. *PeerJ*, 8, e9522.
<https://doi.org/10.7717/peerj.9522>

Warton, D. (2022). *Eco-Stats: Data Analysis in Ecology*. *Springer (Methods in Statistical Ecology)* [Chapter 6]